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### Unmanned aerial image capture platform

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#### Abstract

Methods and systems are disclosed for an unmanned aerial vehicle (UAV) configured to autonomously navigate a physical environment while capturing images of the physical environment. In some embodiments, the motion of the UAV and a subject in the physical environment may be estimated based in part on images of the physical environment captured by the UAV. In response to estimating the motions, image capture by the UAV may be dynamically adjusted to satisfy a specified criterion related to a quality of the image capture.

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References Cited

U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name      | U.S. Cl. | CPC            |
|------------|-------------|--------------------|----------|----------------|
| 5211172    | 12/1992     | McGuane et al.     | N/A      | N/A            |
| 6366311    | 12/2001     | Monroe             | N/A      | N/A            |
| 6744397    | 12/2003     | Hager et al.       | N/A      | N/A            |
| 7363157    | 12/2007     | Hanna et al.       | N/A      | N/A            |
| 7773116    | 12/2009     | Stevens            | N/A      | N/A            |
| 8031175    | 12/2010     | Rigazio et al.     | N/A      | N/A            |
| 8043513    | 12/2010     | Milanovic et al.   | N/A      | N/A            |
| 8301326    | 12/2011     | Malecki et al.     | N/A      | N/A            |
| 8446471    | 12/2012     | Wu                 | 348/148  | G06T<br>3/4038 |
| 8712679    | 12/2013     | Mostofi et al.     | N/A      | N/A            |
| 8958928    | 12/2014     | Seydoux et al.     | N/A      | N/A            |
| 9132913    | 12/2014     | Shapiro et al.     | N/A      | N/A            |
| 9243916    | 12/2015     | Roumeliotis et al. | N/A      | N/A            |
| 9454154    | 12/2015     | Safarik            | N/A      | N/A            |
| 9534917    | 12/2016     | Abuelsaad et al.   | N/A      | N/A            |
| 9588516    | 12/2016     | Gurel et al.       | N/A      | N/A            |
| 9609288    | 12/2016     | Richman et al.     | N/A      | N/A            |
| 9663227    | 12/2016     | Lema et al.        | N/A      | N/A            |
| 9678506    | 12/2016     | Bachrach et al.    | N/A      | N/A            |
| 9690289    | 12/2016     | Yang et al.        | N/A      | N/A            |
| 9738381    | 12/2016     | Loud et al.        | N/A      | N/A            |
| 9739870    | 12/2016     | Beckman et al.     | N/A      | N/A            |
| 9753460    | 12/2016     | Safarik            | N/A      | N/A            |
| 9766074    | 12/2016     | Roumeliotis et al. | N/A      | N/A            |
| 9798322    | 12/2016     | Bachrach et al.    | N/A      | N/A            |
| 9891621    | 12/2017     | Bachrach et al.    | N/A      | N/A            |
| 9930298    | 12/2017     | Bevirt             | N/A      | N/A            |
| 9972212    | 12/2017     | Sperindeo et al.   | N/A      | N/A            |
| 10007265   | 12/2017     | Larsen             | N/A      | N/A            |
| 10033980   | 12/2017     | Boyd et al.        | N/A      | N/A            |

|              |         |                     |            |               |
|--------------|---------|---------------------|------------|---------------|
| 10074183     | 12/2017 | Watson              | N/A        | N/A           |
| 10182225     | 12/2018 | Cui et al.          | N/A        | N/A           |
| 10488860     | 12/2018 | Koch et al.         | N/A        | N/A           |
| 11347217     | 12/2021 | Bachrach et al.     | N/A        | N/A           |
| 11892859     | 12/2023 | Meisenholder et al. | N/A        | N/A           |
| 2007/0078573 | 12/2006 | Ivansson et al.     | N/A        | N/A           |
| 2007/0106473 | 12/2006 | Bodin et al.        | N/A        | N/A           |
| 2008/0033604 | 12/2007 | Margolin            | N/A        | N/A           |
| 2008/0267451 | 12/2007 | Karazi              | N/A        | N/A           |
| 2009/0125223 | 12/2008 | Higgins             | N/A        | N/A           |
| 2009/0157233 | 12/2008 | Kokkeby et al.      | N/A        | N/A           |
| 2009/0228205 | 12/2008 | Ariyur et al.       | N/A        | N/A           |
| 2010/0013860 | 12/2009 | Mandella et al.     | N/A        | N/A           |
| 2010/0084513 | 12/2009 | Gariepy et al.      | N/A        | N/A           |
| 2010/0157055 | 12/2009 | Pechatnikov         | N/A        | N/A           |
| 2010/0191391 | 12/2009 | Zeng                | N/A        | N/A           |
| 2010/0198514 | 12/2009 | Miralles            | N/A        | N/A           |
| 2010/0228414 | 12/2009 | Scheu               | N/A        | N/A           |
| 2010/0250032 | 12/2009 | Gremmert            | N/A        | N/A           |
| 2010/0277587 | 12/2009 | Pechatnikov et al.  | N/A        | N/A           |
| 2010/0302359 | 12/2009 | Adams et al.        | N/A        | N/A           |
| 2010/0305778 | 12/2009 | Dorneich et al.     | N/A        | N/A           |
| 2010/0321470 | 12/2009 | Oshima              | 348/E7.092 | H04N<br>5/772 |
| 2011/0044498 | 12/2010 | Cobb et al.         | N/A        | N/A           |
| 2011/0090399 | 12/2010 | Whitaker et al.     | N/A        | N/A           |
| 2011/0147515 | 12/2010 | Miller et al.       | N/A        | N/A           |
| 2011/0311099 | 12/2010 | Derbanne            | N/A        | N/A           |
| 2012/0114229 | 12/2011 | Zhou                | N/A        | N/A           |
| 2012/0148162 | 12/2011 | Zhang et al.        | N/A        | N/A           |
| 2012/0212406 | 12/2011 | Osterhout et al.    | N/A        | N/A           |
| 2012/0236030 | 12/2011 | Border et al.       | N/A        | N/A           |
| 2013/0030875 | 12/2012 | Lee et al.          | N/A        | N/A           |
| 2013/0127980 | 12/2012 | Haddick et al.      | N/A        | N/A           |
| 2013/0182906 | 12/2012 | Kojo                | 382/103    | G06T 7/579    |
| 2013/0271579 | 12/2012 | Wang                | N/A        | N/A           |
| 2013/0278631 | 12/2012 | Border et al.       | N/A        | N/A           |
| 2013/0317667 | 12/2012 | Kruglick            | N/A        | N/A           |
| 2014/0013364 | 12/2013 | Vondoenhoff et al.  | N/A        | N/A           |
| 2014/0019352 | 12/2013 | Shrivastava         | N/A        | N/A           |
| 2014/0035736 | 12/2013 | Weddle et al.       | N/A        | N/A           |
| 2014/0043436 | 12/2013 | Bell et al.         | N/A        | N/A           |
| 2014/0067160 | 12/2013 | Levien et al.       | N/A        | N/A           |
| 2014/0168461 | 12/2013 | Dani et al.         | N/A        | N/A           |
| 2014/0226024 | 12/2013 | Limbaugh et al.     | N/A        | N/A           |
| 2014/0267777 | 12/2013 | Le Clerc et al.     | N/A        | N/A           |
| 2014/0270743 | 12/2013 | Webb et al.         | N/A        | N/A           |
| 2014/0306866 | 12/2013 | Miller et al.       | N/A        | N/A           |
| 2014/0316616 | 12/2013 | Kugelmass           | N/A        | N/A           |

|              |         |                    |         |               |
|--------------|---------|--------------------|---------|---------------|
| 2014/0316698 | 12/2013 | Roumeliotis et al. | N/A     | N/A           |
| 2014/0324253 | 12/2013 | Duggan et al.      | N/A     | N/A           |
| 2014/0327770 | 12/2013 | Wagreich           | N/A     | N/A           |
| 2014/0336928 | 12/2013 | Scott              | N/A     | N/A           |
| 2014/0371952 | 12/2013 | Ohtomo et al.      | N/A     | N/A           |
| 2015/0022640 | 12/2014 | Metzler et al.     | N/A     | N/A           |
| 2015/0027044 | 12/2014 | Redden             | N/A     | N/A           |
| 2015/0042800 | 12/2014 | Choi               | 348/148 | H04N<br>7/181 |
| 2015/0062339 | 12/2014 | Ostrom             | N/A     | N/A           |
| 2015/0153436 | 12/2014 | Benson             | N/A     | N/A           |
| 2015/0158587 | 12/2014 | Patrick et al.     | N/A     | N/A           |
| 2015/0160658 | 12/2014 | Reedman et al.     | N/A     | N/A           |
| 2015/0201180 | 12/2014 | Mourikis et al.    | N/A     | N/A           |
| 2015/0230150 | 12/2014 | Wang et al.        | N/A     | N/A           |
| 2015/0242972 | 12/2014 | Lemmey et al.      | N/A     | N/A           |
| 2015/0259078 | 12/2014 | Filipovic et al.   | N/A     | N/A           |
| 2015/0310603 | 12/2014 | Moraites et al.    | N/A     | N/A           |
| 2015/0312774 | 12/2014 | Lau                | N/A     | N/A           |
| 2015/0341540 | 12/2014 | Kim et al.         | N/A     | N/A           |
| 2015/0346915 | 12/2014 | Kondekar et al.    | N/A     | N/A           |
| 2015/0370250 | 12/2014 | Bachrach et al.    | N/A     | N/A           |
| 2016/0018822 | 12/2015 | Nevdahs et al.     | N/A     | N/A           |
| 2016/0035224 | 12/2015 | Yang et al.        | N/A     | N/A           |
| 2016/0041266 | 12/2015 | Smits              | N/A     | N/A           |
| 2016/0050840 | 12/2015 | Sauder et al.      | N/A     | N/A           |
| 2016/0054737 | 12/2015 | Soll et al.        | N/A     | N/A           |
| 2016/0068267 | 12/2015 | Liu et al.         | N/A     | N/A           |
| 2016/0070265 | 12/2015 | Liu et al.         | N/A     | N/A           |
| 2016/0110618 | 12/2015 | Oba                | 348/148 | G06F 18/22    |
| 2016/0122038 | 12/2015 | Fleischman et al.  | N/A     | N/A           |
| 2016/0129999 | 12/2015 | Mays               | N/A     | N/A           |
| 2016/0139596 | 12/2015 | Na et al.          | N/A     | N/A           |
| 2016/0140729 | 12/2015 | Soatto et al.      | N/A     | N/A           |
| 2016/0144943 | 12/2015 | Cheng et al.       | N/A     | N/A           |
| 2016/0176538 | 12/2015 | Bekanich           | N/A     | N/A           |
| 2016/0232423 | 12/2015 | Zhong et al.       | N/A     | N/A           |
| 2016/0267325 | 12/2015 | Sundaresan et al.  | N/A     | N/A           |
| 2016/0280397 | 12/2015 | Christ et al.      | N/A     | N/A           |
| 2016/0299504 | 12/2015 | Hsiao              | N/A     | N/A           |
| 2016/0304198 | 12/2015 | Jourdan            | N/A     | N/A           |
| 2016/0327950 | 12/2015 | Bachrach et al.    | N/A     | N/A           |
| 2016/0344981 | 12/2015 | Lunt               | N/A     | N/A           |
| 2017/0008521 | 12/2016 | Braunstein et al.  | N/A     | N/A           |
| 2017/0010623 | 12/2016 | Tang et al.        | N/A     | N/A           |
| 2017/0023937 | 12/2016 | Loianno et al.     | N/A     | N/A           |
| 2017/0024877 | 12/2016 | Versace et al.     | N/A     | N/A           |
| 2017/0031032 | 12/2016 | Garin et al.       | N/A     | N/A           |
| 2017/0039764 | 12/2016 | Hu et al.          | N/A     | N/A           |
| 2017/0039859 | 12/2016 | Hu et al.          | N/A     | N/A           |

|              |         |                    |     |               |
|--------------|---------|--------------------|-----|---------------|
| 2017/0066135 | 12/2016 | Cohen et al.       | N/A | N/A           |
| 2017/0116776 | 12/2016 | Aughey             | N/A | N/A           |
| 2017/0140542 | 12/2016 | Hodohara           | N/A | H04N<br>25/61 |
| 2017/0180729 | 12/2016 | Wu                 | N/A | N/A           |
| 2017/0180754 | 12/2016 | Wu et al.          | N/A | N/A           |
| 2017/0192418 | 12/2016 | Bethke et al.      | N/A | N/A           |
| 2017/0201714 | 12/2016 | Kim et al.         | N/A | N/A           |
| 2017/0210486 | 12/2016 | O'Brien et al.     | N/A | N/A           |
| 2017/0219347 | 12/2016 | Veto               | N/A | N/A           |
| 2017/0227656 | 12/2016 | Niesen et al.      | N/A | N/A           |
| 2017/0275023 | 12/2016 | Harris et al.      | N/A | N/A           |
| 2017/0278014 | 12/2016 | Lessmann et al.    | N/A | N/A           |
| 2017/0294010 | 12/2016 | Shen et al.        | N/A | N/A           |
| 2017/0301109 | 12/2016 | Chan et al.        | N/A | N/A           |
| 2017/0305546 | 12/2016 | Ni et al.          | N/A | N/A           |
| 2017/0313416 | 12/2016 | Mishra et al.      | N/A | N/A           |
| 2017/0313441 | 12/2016 | Tsai               | N/A | N/A           |
| 2017/0314926 | 12/2016 | Royster et al.     | N/A | N/A           |
| 2017/0314927 | 12/2016 | Royster et al.     | N/A | N/A           |
| 2017/0329324 | 12/2016 | Bachrach et al.    | N/A | N/A           |
| 2017/0341776 | 12/2016 | McClure et al.     | N/A | N/A           |
| 2017/0351933 | 12/2016 | Bleiweiss          | N/A | N/A           |
| 2017/0357858 | 12/2016 | Mendonca et al.    | N/A | N/A           |
| 2017/0358099 | 12/2016 | Harris et al.      | N/A | N/A           |
| 2017/0359515 | 12/2016 | Harris et al.      | N/A | N/A           |
| 2017/0359943 | 12/2016 | Calleija et al.    | N/A | N/A           |
| 2017/0371353 | 12/2016 | Millinger, III     | N/A | N/A           |
| 2018/0046187 | 12/2017 | Martirosyan et al. | N/A | N/A           |
| 2018/0074524 | 12/2017 | Yamasaki           | N/A | N/A           |
| 2018/0095459 | 12/2017 | Bachrach et al.    | N/A | N/A           |
| 2018/0129278 | 12/2017 | Luchinskiy         | N/A | N/A           |
| 2018/0157255 | 12/2017 | Halverson et al.   | N/A | N/A           |
| 2018/0201272 | 12/2017 | Takeda             | N/A | N/A           |
| 2018/0234496 | 12/2017 | Ratias             | N/A | N/A           |
| 2018/0246507 | 12/2017 | Bachrach et al.    | N/A | N/A           |
| 2019/0011921 | 12/2018 | Wang et al.        | N/A | N/A           |
| 2019/0027036 | 12/2018 | Mishina et al.     | N/A | N/A           |
| 2019/0035278 | 12/2018 | Mishina et al.     | N/A | N/A           |
| 2019/0149735 | 12/2018 | Harris et al.      | N/A | N/A           |
| 2019/0204824 | 12/2018 | Micros             | N/A | N/A           |

#### FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 2010219335  | 12/2009          | AU      | N/A |
| 2013006210  | 12/2012          | WO      | N/A |
| 2015/105886 | 12/2014          | WO      | N/A |
| 2016/086379 | 12/2015          | WO      | N/A |

#### OTHER PUBLICATIONS

Aguiar, Antonio Pedro et al., “Logic-Based Switching Control for Trajectory-Tracking and Path-Following of Underactuated Autonomous Vehicles With Parametric Modeling Uncertainty,” Proceeding of the 2004 American Control Conference, pp. 3004-3010, Jun. 30-Jul. 2, 2004. cited by applicant

Ataei, Mansour et al., “Three-Dimensional Optimal Path Planning For Waypoint Guidance of an Autonomous Underwater Vehicle,” Robotics and Autonomous Systems, vol. 67, pp. 23-32, 2015, available online Oct. 24, 2014. cited by applicant

Brake, Nicholas J., Master's Thesis for “Control System Development for Small UAV Gimbal,” 113 pages, Aug. 2012. cited by applicant

Gu, Feng et al., “Active Persistent Localization of A Three-Dimensional Moving Target Under Set-Membership Uncertainty Description Through Cooperation of Multiple Mobile Robots,” IEEE Transactions on Industrial Electronics, vol. 62, No. 8, pp. 4958-4971, Aug. 2015, initial publication Feb. 19, 2015. cited by applicant

Horenstein, Henry, “Black & White Photography—A Basic Manual,” Third Revised Edition, Chapter 6, p. 94, 2005. cited by applicant

Kakvand, P. et al., “Smart On-Board UAV System: Using Computer Vision System to Find a Movable and Stationary Target,” IEEE, pp. 694-699, 2015. cited by applicant

Kalnins, L. M., “Coordinate Systems,” 5 pages, Mar. 2009. cited by applicant

Kaminer, Isaac et al., “Trajectory Tracking for Autonomous Vehicles: An Integrated Approach to Guidance and Control,” Journal of Guidance, Control, and Dynamics, vol. 21, No. 1, pp. 29-38, Jan.-Feb. 1998. cited by applicant

Kim, Seungkeun et al., “Three Dimensional Optimum Controller for Multiple UAV Formation Flight Using Behavior-Based Decentralized Approach,” International Conference on Control, Automation and Systems, pp. 1387-1392, Oct. 17-20, 2007. cited by applicant

Mansfield, Katrina et al., “Unmanned Aerial Vehicle Smart Device Ground Control Station Cyber Security Threat Model,” IEEE, pp. 722-728, 2013. cited by applicant

O'Reilly, O. M., “Engineering Dynamics: A Primer,” Chapter 2, pp. 17-25, 2010. cited by applicant

Quigley, Morgan et al., “Target Acquisition, Localization, and Surveillance Using a Fixed-Wing Mini-UAV and Gimbaled Camera,” IEEE, pp. 2600-2605, 2005. cited by applicant

Santana, Lucas Vago et al., “Outdoor Waypoint Navigation With The AR.Drone Quadrotor,” IEEE, pp. 303-311, 2015. cited by applicant

International Application No. PCT/US2016/027921, International Search Report, Written Opinion, 12 pages, Jul. 15, 2016. cited by applicant

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/899,459, entitled “UNMANNED AERIAL IMAGE CAPTURE PLATFORM,” filed Aug. 30, 2022; which is a continuation of U.S. patent application Ser. No. 17/388,828, entitled “UNMANNED AERIAL IMAGE CAPTURE PLATFORM,” filed Jul. 29, 2021; which is a continuation of U.S. patent application Ser. No. 16/723,549, entitled “UNMANNED AERIAL IMAGE CAPTURE PLATFORM,” filed Dec. 20, 2019; which is a continuation of U.S. patent application Ser. No. 15/235,513, entitled “UNMANNED AERIAL IMAGE CAPTURE PLATFORM,” filed Aug. 12, 2016, each of which is hereby incorporated by

reference in its entirety for all purposes. This application is therefore entitled to a priority date of Aug. 12, 2016.

## TECHNICAL FIELD

(1) The present disclosure relates generally to methods and systems for the control of unmanned aerial vehicles (UAV) used as platforms for the capture of images (including video).

## BACKGROUND

(2) Unmanned aerial vehicles (UAV) are increasingly being used as platforms for taking images and video from the air. A number of UAV systems are currently available that provide for image and video capture and remote control from a device on the ground. However, currently available systems require piloting using direct control of the UAV similar to other fixed wing or rotor-craft. In other words, available systems require control by directly adjusting the pitch, roll, yaw, and power of the UAV, for example using common control inputs such as a joystick and throttle control. While effective to a degree, such control systems require expertise on the part of the remote pilot and are prone to crashes caused by pilot error.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is illustration of an example unmanned aerial vehicle (“UAV”).
- (2) FIG. 2 is a conceptual diagram of an example localization and navigation system for guiding navigation and image capture by a UAV.
- (3) FIG. 3 is a conceptual diagram of an example system for estimating the position and/or orientation of a UAV using a network of phased array wireless transceivers.
- (4) FIG. 4 is a conceptual diagram of an example system for passive localization of a subject tracked by a UAV.
- (5) FIGS. 5A-5B illustrate example techniques for estimating the position and/or orientation of objects using images captured by a UAV.
- (6) FIG. 6 illustrates an example arrangement of image capture devices coupled to a UAV.
- (7) FIG. 7A illustrates an example technique for defining the motion of a UAV relative to a point of reference using a cylindrical polar coordinate system.
- (8) FIG. 7B illustrates an example technique for defining the motion of a UAV relative to a point of reference using a spherical polar coordinate system.
- (9) FIG. 8A illustrates an example scenario including a UAV in flight through a physical environment while capturing images of a subject.
- (10) FIG. 8B is a flow chart describing an example process for dynamically adjusting image capture of a subject by a UAV.
- (11) FIG. 9 illustrates an example scenario including a UAV adjusting image capture of a subject to avoid a collision with another object.
- (12) FIG. 10 illustrates an example scenario including a UAV adjusting image capture to keep a subject in view.
- (13) FIG. 11 illustrates an example scenario including a UAV adjusting image capture of a subject according to a predefined flight path.
- (14) FIG. 12 illustrates an example scenario including a UAV adjusting image capture of a subject to avoid backlighting.
- (15) FIG. 13 is a diagram of an example UAV system.
- (16) FIG. 14 is a diagram of a computing system.

### DETAILED DESCRIPTION

#### Overview

- (17) FIG. 1 is an illustration of an unmanned aerial vehicle (“UAV”) **100** operating as an automated

aerial platform for capturing images. As shown in FIG. 1, in some embodiments, UAV 100 may be a rotor-based aircraft (e.g., a “quadcopter”). The UAV 100 as shown in FIG. 1 may include propulsion and control actuators 110 (e.g. powered rotors or aerodynamics control surfaces) for maintaining controlled flight, sensors for automated navigation and flight control 112 (e.g. an array of multiple image capture devices—described in more detail herein), a gimbaled image capture device 114 for capturing images (including video), other sensors (not shown) (e.g. for capturing audio), and means (not shown) for communicating with other devices (e.g. a mobile device 104), for example via a wireless connection 116. The UAV 100 shown in FIG. 1 is according to an example embodiment. A UAV 100 in accordance with the present teachings may include more or fewer components than as shown. An example UAV system that may be part of UAV 100 is described later with respect to FIG. 13.

(18) In FIG. 1, UAV 100 is shown capturing images of a human subject 102 that in some embodiments may be a user (i.e. in communication with UAV 100 via a mobile device 104). However, a communication link with another device (e.g. mobile device 104) and control by a human user is not necessary. UAV 100 may autonomously (i.e. without direct human control) navigate the physical environment, for example by applying a process of visual inertial odometry using images captured by an array of multiple image capture devices. While in autonomous flight, UAV 100 can capture images of one or more subjects in the physical environment using the same array of image capture devices and/or the separate gimbaled image capture device 114. In this sense, UAV 100 may generally be conceptualized as an autonomous aerial camera rather than as a vehicle with an attached camera, and may therefore represent a paradigm shift in which cameras are understood.

(19) To improve the quality of image capture (objectively and/or subjectively), one or more criteria may be specified that define how UAV 100 is to respond to given conditions while autonomously capturing images over a physical environment. In other words, to satisfy the specified one or more criteria, UAV 100 may be configured to automatically adjust image capture, which may in some cases include adjusting its flight path. As an illustrative example, consider an example criterion that states that while tracking and capturing images of a subject in motion, the UAV 100 is to always (or at least within a threshold tolerance) maintain a clear line of sight with the subject. In other words, it is not enough to stay within a maximum separation distance. If the line of sight with the subject becomes obstructed by another object in the physical environment, the UAV may automatically adjust its flight path to alleviate the obstruction. The particular maneuver required in any given situation depends on the geometric configuration of the subject and the UAV within the physical environment. As an illustrative example, consider a UAV 100 tracking a human subject in motion. As the human subject moves under a tree, the view from the UAV 100 located overhead becomes obstructed by the leaves of the tree. To satisfy the specified criterion (of maintaining clear line of sight) a processing unit (located on board the UAV or remotely and in communication with the UAV) may generate commands configured to adjust image capture, for example, by causing the UAV 100 to reduce altitude below the level of the leaves to alleviate the obstruction in the view.

(20) Flying Digital Assistant—Localization and Navigation

(21) FIG. 2 is a high-level illustration of a localization and navigation system 200, according to some embodiments, for guiding navigation and image capture by an UAV 100. According to some embodiments, a relative position and/or orientation of the UAV 100, and or a relative position and/or orientation of a subject 102 may be determined using one or more of the subsystems illustrated in FIG. 2. It shall be understood that by tracking relative positions and/or orientations over time (continuously or at regular or irregular time intervals (i.e. continually)), the motions (e.g. velocity, acceleration, etc.) of UAV 100 and one or more subjects through a physical environment may be estimated. Accordingly, any systems described herein for determining position and/or orientation may similarly be employed for estimating motion. According to the present teaching localization a system 200 may include a UAV 100, a global positioning system (GPS) comprising



multiple GPS satellites **202**, a cellular system comprising multiple cellular antennae **204** (with access to sources of localization data **206**), a Wi-Fi system comprising multiple Wi-Fi routers **208** (with access to sources of localization data **206**), and a portable multifunction device **104** operated by a user (in this example human subject **102**).

(22) The UAV **100** may comprise components including, but not limited to, an inertial measurement unit (IMU), a GPS receiver, multiple RF receivers and/or transceivers (e.g. cellular LTE, Wi-Fi), and one or more image capture devices. For example, an image capture device may be used to determine position and/or pose through the use of computer vision techniques and or optics-based collision detection and range finding. This is illustrated conceptually in FIG. **2** by the dotted line **214**. These components are described in more detail in the section titled “Unmanned Aerial Vehicle—Example System” and with reference to FIG. **13**.

(23) Mobile device **104** may be any type of portable computing device. For example, mobile device **104** may include a notebook, a laptop computer, a handheld computer, a palmtop computer, a cell phone, a PDA, a smart phone (e.g., iPhone™, etc.), a tablet (e.g., iPad™, etc), a hand held gaming device (e.g., Sony PSP™, etc.), a smart watch (e.g., Apple Watch™, etc.), an augmented reality device (e.g., Google Glass™, etc.), a virtual reality device (e.g. Oculus Rift™, etc.) or any other portable computing device. Further, mobile device **104** may include any of the components described with respect systems **1300** or **1400** as shown in FIGS. **13** and **14** (respectively), including, but not limited to, an inertial measurement unit (IMU), a GPS receiver, multiple RF receivers and/or transceivers (e.g. cellular LTE, Wi-Fi), and an image capture device. As used in this disclosure, the term “system” may refer to a computing system operating as part of and/or in conjunction with any of the computing systems described with respect to FIGS. **13** and **14**. In some embodiments, methods described herein may be performed by a computing system that is not part of UAV **100** or a mobile device **104**, for example a general computing system illustrated with respect to FIG. **14**.

(24) As mentioned earlier, a relative position and/or orientation of the UAV **100**, a relative position and/or orientation of the subject **102**, and/or a relative position and/or pose of a mobile device **104** operated by a user may be determined using one or more of the subsystems illustrated in FIG. **2**. For example, using only the GPS system **202**, a position on the globe may be determined for any device comprising a GPS receiver (e.g. the UAV **100** and/or the mobile device **104**). While GPS by itself in certain implementations may provide highly accurate global positioning it is generally is not capable of providing accurate information regarding orientation. Instead a technique of multiple inputs and multiple outputs (“MIMO”) (as illustrated in FIG. **2**) may be used for localization, potentially in conjunction with other localization subsystems.

(25) Consider the example based on the illustration in FIG. **2**; a user (human subject **102**) is utilizing an autonomous UAV **100** via a mobile device **104** to film herself overhead. In order navigate the UAV **100** and inform the tracking by an image capture device of the subject **102** (in this case the user), a relative position and orientation of the UAV **100** relative to the mobile device **104** (or any other point of reference) may be necessary.

(26) According to the present teachings a relative position between the UAV **100** and the mobile device **104** may be determined using a GPS system to determine a global position of the UAV **100**, a global position of the mobile device **104** and compare the two.

(27) Similarly, using an array of cellular and or/Wi-fi antennae, a position relative to the known locations of antennae may be determined for both the UAV **100** and mobile device **104** using known positioning techniques. Some known positioning techniques include those based on signal trilateration, for example round trip time of arrival (RTT) in which a signal is sent and received by a signal transceiver and distance is calculated based on the elapsed time, received signal strength (RSS) in which the power levels of the transmitted signal and the received signals are analyzed and a distance determined based on a known propagation loss. Other known positioning techniques include those based on signal triangulation, for example angle of arrival (AoA) in which angles of

arriving signals are determined and through applied geometry a position determined. Current Wi-Fi standards, such as 802.11ac, allow for RF signal beamforming (i.e. directional signal transmission using phased-shifted antenna arrays) from transmitting Wi-Fi routers. Beamforming may be accomplished through the transmission of RF signals at different phases from spatially distributed antennas (a “phased antenna array”) such that constructive interference may occur at certain angles while destructive interference may occur at others, thereby resulting in a targeted directional RF signal field. Such a targeted field is illustrated conceptually in FIG. 2 by dotted lines 212 emanating from WiFi routers 210.

(28) As illustrated in FIG. 3, a UAV 100 and/or mobile device 104 may include a phased array of WiFi antenna and a relative position and/or pose may be calculated without the necessity for external existing Wi-Fi routers. According to some embodiments, the UAV 100 and/or mobile device 104 may transmit and/or receive a beamformed RF signal via a phased antenna array. The UAV 100 and/or mobile device 104 may then detect the phase differences and power levels of the respective incoming signals and calculate an AoA for the incoming signals. For example according to FIG. 3, the mobile device 104 may determine an AoA of  $\theta_1$  for the RF signals 302 transmitted by the UAV 100. Similarly, the UAV 100 may determine an AoA of  $\theta_2$  for the RF signals 304 transmitted by the mobile device 104. This AoA information may then be incorporated with information gathered by an IMU on the UAV 100 and/or mobile device 104 (as well as other positioning data as described earlier) in order to infer a relative position and/or orientation between the UAV 100 and the mobile device 104.

(29) According to some embodiments, an array of Wi-Fi transmitters and signal monitors may be utilized for device-free passive localization of objects that are not transmitting signals (e.g. a human subject not carrying a mobile device). FIG. 4 illustrates an example system 400 for device-free passive localization of subject (e.g. a human subject 102). In this example a human subject 102 passes through a network of Wi-Fi transmitters 408 transmitting RF signals. The signal monitors 410 (e.g. standard wireless sniffers) may detect changes in the characteristics of the RF signals received from the Wi-Fi transmitters 408 caused by interference as the human subject 102 passes through the signal field. Using localization algorithms, such changes in the RF signal field may be correlated to the presence of the subject 102, its type, its orientation and its location. Also, according to FIG. 4., information gathered by device-free passive localization system 400 may be fed wirelessly (e.g. via Wi-Fi connection 430) for to a nearby UAV 100 in order to inform its tracking of the human subject 102.

(30) According to some embodiments an inertial measurement unit (IMU) may be used to determine relative position and/or orientation. An IMU is a device that measures a vehicle's angular velocity and linear acceleration. These measurements can be fused with other sources of information (e.g. those discussed above) to accurately infer velocity, orientation, and sensor calibrations. As described herein, a UAV 100 and/or mobile device 104 may include one or more IMUs. Using a method commonly referred to as “dead reckoning” an IMU (or associated systems) may calculate and track a predicted a current position based on a previously known position(s) using measured accelerations and the time elapsed from the previously known position(s). While effective to an extent, the accuracy achieved through dead reckoning based on measurements from an IMU quickly degrades due to the cumulative effect of errors in each predicted current position. Errors are further compounded by the fact that each predicted position is based on a calculated integral of the measured velocity. To counter such effects, an embodiment utilizing localization using an IMU may include localization data from other sources (e.g. the GPS, Wi-Fi, and cellular systems described above) to continually update the last known position and/or orientation of the object. Further, a nonlinear estimation algorithm (one embodiment being an “extended Kalman filter”) may be applied to a series of measured positions and/or orientations to produce a real-time optimized prediction of the current position and/or orientation based on assumed uncertainties in the observed data. Kalman filters are commonly applied in the area of aircraft navigation, guidance,

and controls.

(31) According to some embodiments, computer vision may be used to determine a relative position and/or orientation of a UAV **100**, mobile device **104**, and or any other object. The term, “computer vision” in this context may generally refer to the acquiring, processing, analyzing and “understanding” of captured images. Consider again the localization system illustrated in FIG. 2. According to some embodiments, a UAV **100** may include an image capture device and computer vision capabilities. In this example, UAV **100** may be programed to track a user **102** (or other physical object). Using computer vision, a UAV **100** may recognize the captured image as the subject and may use the recognition information to perform aerial maneuvers by the UAV **100** to keep the subject in view, and/or may make adjustments to an image stabilization system (e.g. a gimbaled image capture device) to keep the subject in view.

(32) Relative position and/or orientation may be determined through computer vision using a number of methods. For example, in some embodiments, raw image data received from one or more image capture devices (onboard or remote from the UAV **100**) may be received and processed to correct for certain variables (e.g. differences in camera orientation and/or intrinsic parameters (e.g. lens variations)). According to some embodiments, an image capture device of the UAV **100** may include two or more cameras, for example an array of multiple cameras that provide an unobstructed view around the UAV **100**. By comparing the captured image from two or more vantage points (e.g. at different time steps from an image capture device in motion), a system employing computer vision may calculate estimates for the relative position and/or orientation of the vehicle on which the image capture device is mounted (e.g. UAV **100**) and/or of a captured object in the physical environment (e.g. the subject). With the calculated position and/or orientation data for the UAV **100** (e.g., data from GPS, WiFi, Cellular, and/or IMU, as discussed above) a relative position and/or orientation may be determined between the UAV **100** and the captured physical object.

(33) According to some embodiments, an image capture device of UAV **100** may be a single camera (i.e. a non-stereoscopic camera). Here, computer vision algorithms may identify the presence of an object and identify the object as belonging to a known type with particular dimensions. In such embodiments, an object may be identified by comparing the captured image to stored two-dimensional (2D) and/or three-dimensional (3D) appearance models. For example, through computer vision, the subject **102** may be identified as an adult male human. In some embodiments the 2D and/or 3D appearance models may be represented as a trained neural network that utilizes deep learning to classify objects in images according to detected patterns. With this recognition data, as well as other position and/or orientation data for the UAV **100** (e.g. data from GPS, WiFi, Cellular, and/or IMU, as discussed above), UAV **100** may estimate a relative position and/or orientation of the subject **102**.

(34) According to some embodiments, computer vision may be used along with measurements from an IMU (or accelerometer(s) or gyroscope(s)) within the UAV **100** and/or mobile device **104** carried by a user (e.g. human subject **102**) as illustrated in FIG. 5A-5B. FIG. 5A shows a simplified diagram that illustrates how sensor data gathered by an IMU at a mobile device **104** may be applied to sensor data gathered by an image capture device at UAV **100** to determine position and/or orientation data of a physical object (e.g. a user **102**). Outline **550** represents the two-dimensional image captured field of view at UAV **100**. As shown in FIG. 5A, the field of view includes the image of a physical object (here user **102**) moving from one position to another. From its vantage point, UAV **100** may determine a distance A traveled across the image capture field of view. The mobile device **104**, carried by user **102**, may determine an actual distance B traveled by the user **102** based on measurements by internal sensors (e.g. the IMU) and an elapsed time. The UAV **100** may then receive the sensor data and/or the distance B calculation from mobile device **104** (e.g., via wireless RF signal). Correlating the difference between the observed distance A and the received distance B, UAV **100** may determine a distance D between UAV **100** and the physical

object (user **102**). With the calculated distance as well as other position and/or orientation data for the UAV **100** (e.g. data from GPS, WiFi, Cellular, and/or IMU, as discussed above) a relative position and/or orientation may be determined between the UAV **100** and the physical object (e.g. user **102**).

(35) Alternatively, estimations for the position and/or orientation of either the UAV **100** or mobile device **104** may be made using a process generally referred to as “visual inertial odometry” or “visual odometry.” FIG. 5B illustrates the working concept behind visual odometry at a high level. A plurality of images are captured in sequence as an image capture device moves through space. Due to the movement of the camera, the images captured of the surrounding space change from frame to frame. In FIG. 5B, this is illustrated by initial image capture field of view **552** and a subsequent image capture field of view **554** captured as the camera has moved from a first position and orientation to a second position and orientation over an elapsed time. In both images, the camera may capture real world physical objects, for example, the house **580** and/or the human subject **102**. Computer vision techniques are applied to the sequence of images to detect and match features of physical objects captured in the field of view of the camera. For example, a system employing computer vision may search for correspondences in the pixels of digital images that have overlapping fields of view (FOV). The correspondences may be identified using a number of different methods such as correlation-based and feature-based methods. As shown in, in FIG. 5B, features such as the head of a human subject **102** or the corner of the chimney on the house **580** can be identified, matched, and thereby tracked. By incorporating sensor data from an IMU (or accelerometer(s) or gyroscope(s)) associated with the camera to the tracked features of the image capture, estimations may be made for the position and/or orientation of the camera over time. Further, these estimates can be used to calibrate the localization systems, for example through estimating differences in camera orientation and/or intrinsic parameters (e.g. lens variations)) or IMU biases and/or orientation. Visual inertial odometry may be applied at both the UAV **100** and mobile device **104** to calculate the position and/or orientation of both systems. Further, by communicating the estimates between the systems (e.g. via a Wi-Fi connection) estimates may be calculated for the respective positions and/or orientations relative to each other. As previously mentioned, position, orientation, and motion estimation based in part on sensor data from an on board IMU may introduce error propagation issues. As previously stated, optimization techniques may be applied to position, orientation, and motion estimations to counter such uncertainties. In some embodiments, a nonlinear estimation algorithm (one embodiment being an “extended Kalman filter”) may be applied to a series of measured positions and/or orientations to produce a real-time optimized prediction of the current position and/or orientation based on assumed uncertainties in the observed data. Such estimation algorithms can be similarly applied to produce smooth motion estimations.

(36) In some embodiments, systems in accordance with the present teachings may simultaneously generate a 3D map of the surrounding physical environment while estimating the relative positions and/or orientations of the UAV **100** and/or subject within the physical environment. This is sometimes referred to simultaneous localization and mapping (“SLAM”). In such embodiments, using computer vision processing, a system in accordance with the present teaching can search for dense correspondence between images with overlapping FOV (e.g. images taken during sequential time steps and/or stereoscopic images taken at the same timestep). The system can then use the dense correspondences to estimate a depth or distance to each pixel represented in each image. These depth estimates can then be used to continually update a generated 3D model of the physical environment taking into account motion estimates for the image capture device (i.e. UAV **100**) through the physical environment.

(37) According to some embodiments, computer vision may include sensing technologies other than image capture devices (i.e. cameras) such as laser illuminated detection and ranging (LIDAR or Lidar). For example, a UAV **100** equipped with LIDAR may emit one or more laser beams in a

continuous scan up to 360 degrees around the UAV **100**. Light received by the UAV **100** as the laser beams reflect off physical objects in the surrounding physical world may be analyzed to construct a real time 3D computer model of the surrounding physical world. Depth sensing through the use of LIDAR may in some embodiments augment depth sensing through pixel correspondence as described earlier. Such 3D models may be analyzed to identify particular physical objects (e.g. subject **102**) in the physical environment for tracking. Further, images captured by cameras (e.g., as described earlier) may be combined with the laser constructed 3D models to form textured 3D models that may be further analyzed in real time or near real time for physical object recognition (e.g. by using computer vision algorithms).

(38) The computer vision-aided localization and navigation system described above may calculate the position and/or pose of features in the physical world in addition to the position and/or pose of the UAV **100** and/or mobile device **104**. The position of these features may then be fed into the navigation system such that motion trajectories may be planned that avoid obstacles. In addition, in some embodiments, the visual navigation algorithms may incorporate data from proximity sensors (e.g. electromagnetic, acoustic, and/or optics based) to estimate obstacle position with more accuracy. Further refinement may be possible with the use of stereoscopic computer vision with multiple cameras, as described earlier.

(39) According to some embodiments, the previously described relative position and/or orientation calculations may be performed by a UAV **100**, mobile device **104**, other remote computing device(s) (not shown in the figures), or any combination thereof.

(40) The localization system **200** of FIG. 2 (including all of the associated subsystems as previously described) is only one example of a system for localization and navigation. Localization system **200** may have more or fewer components than shown, may combine two or more components, or a may have a different configuration or arrangement of the components. Some of the various components shown in FIGS. 2 through 4 may be implemented in hardware, software or a combination of both hardware and software, including one or more signal processing and/or application specific integrated circuits.

(41) Flying Digital Assistant—Array of Image Capture Devices

(42) According to some embodiments, UAV **100** may comprise multiple high-resolution image capture devices **602** (e.g. cameras) with spatial offsets from each other, thereby providing the capability to capture an unobstructed view of the physical environment surrounding UAV **100**. In some embodiments, image capture devices **602** may be arranged to provide a full 360 degree view around UAV **100**, as illustrated in FIG. 6. However, a full 360 degree view may not be necessary in all embodiments. In some embodiments, the image capture devices **602** may be arranged such that at least two cameras are provided with overlapping fields of view, thereby allowing for stereoscopic (i.e. 3D) image/video capture and depth recovery (e.g. through computer vision algorithms) at multiple angles around UAV **100**. For example, FIG. 6 shows a high-level illustration of the concept of multiple image capture devices **602** mounted to UAV **100** with overlapping fields of view as represented by the dotted lines. FIG. 6 is provided to illustrate the concept, but does not indicate a particular configuration or geometry as a limitation. According to some embodiments, a UAV in accordance with the present teachings may include more or fewer image capture devices **602**. For example, in some embodiments, the individual fields of view of any given image capture device may be expanded through the use of a “fisheye” lens, thereby reducing the total number of image capture devices needed to provide a 360 degree view around UAV **100**.

(43) According to some embodiments the position and orientation of each camera may be calibrated to an onboard inertial measurement unit (IMU) by fusing data from the cameras and IMUs in a visual inertial odometry framework.

(44) According to some embodiments, a monocular navigation algorithm may be run for each camera paired with an on-board IMU and as the relative position and orientation calibration is

dialled in, stereo correspondence may be performed on observed primitives representing a pair of corresponding image features captured by a pair of cameras in order to provide a more robust estimation of distance to the objects.

(45) Hybrid Mechanical/Digital Gimbal

(46) In some embodiments, UAV **100** includes an image capture adjustment and stabilization system. Capturing images (including video) from a vehicle in motion (such as from a UAV **100**) may lead to quality issues such as blur, shake, and disorientation. Image stabilization may generally refer to techniques used to counter these effects and produce a clear stable image even when captured by a vehicle in motion.

(47) A number of techniques and devices for image stabilization are currently known. For example, a multi-axis mechanical gimbal device may, through the use of gyroscopes and mechanical actuators along two or more axis, physically stabilize an image capturing device (e.g. a camera) coupled to a mobile platform. An example of a multi-axis gimbal currently available is the Freefly MoVI™. While effective in certain implementations, multi-axis mechanical gimbals may add significant mechanical and systems complexity as well as weight to a UAV **100**. Alternatively, captured digital images may be digitally “stabilized” using digital image processing to manipulate the image. For example, Parrot™ offers a drone with a motionless 180 degree camera with a fisheye lens. Using post processing and crop filters may result in a “stabilized” image. While effective in certain implementations, full digital image stabilization may reduce image quality due to image sensor resolution limits, and in the case of using crop filters may require capturing more data than is necessary.

(48) Instead a UAV **100**, according to some embodiments, may include a hybrid approach comprising mechanical gimbals providing freedom of motion along one or more axes along with real-time image processing (herein referred to as a “digital gimbal”). For example, a single axis mechanical gimbal capable of adjusting the orientation of an image capture device in conjunction with the yaw control of the UAV **100** and digital image processing may produce a full range of image capture from looking straight down from the UAV **100** to the ground to looking straight up from the UAV **100** to the sky while minimizing the mechanical complexity of the stabilization system.

(49) According to some embodiments, a single axis mechanical gimbal, as part of a hybrid approach described above, would adjust the pitch of the image capture device. Adjusting pitch as opposed to roll or yaw, would allow for overall camera range of motion where the UAV **100** is implemented as a rotary vehicle, for example a quadcopter (see e.g. discussion in section titled “Unmanned Aerial Vehicle—Example System” for additional information). This has to do with the way in which the flight of a quadcopter is controlled. Generally, a quadcopter is controlled by varying the orientation of its vertical axis. In other words, in a hover the quadcopter's vertical axis is perpendicular to the ground. In order to move left or right, forwards or backwards, the angular velocity of the four rotors are adjusted, and the quadcopter tilts in the direction that it intends to move. This method of control leaves the quadcopter free to determine yaw, thus effectively “gimbaling” the yaw axis. While using yaw to point at a desired subject may be difficult for a human pilot, it can be accomplished by the UAV's **100** flight control system and the localization techniques described herein. Accordingly, utilizing a pitch gimbal gives maximum possible view range of motion since the yaw of the image capture device is easily controlled by adjusting the yaw of the quadcopter itself and the roll of the image capture device is easily controlled through digital image processing, for example simple image rotation transforms. The hybrid mechanical digital gimbal system described above has been described with a single axis mechanical gimbal, however it shall be appreciated that a hybrid mechanical digital gimbal system for image stabilization and tracking may include mechanical actuation on more than one axis. For example, in an embodiment, a mechanical gimbal is utilized to adjust the pitch and roll of an on-board image capture device with adjustments in yaw accomplished by digitally processing the captured images.

(50) Changes in Position and/or Orientation Relative to a Point of Reference

(51) According to some embodiments, the UAV **100** may maneuver according to an absolute fixed coordinate system. In other words, user inputs and gestures may correspond with an instruction to move to an absolute point in space. The UAV **100** may also maneuver according to a coordinate system relative to a “point of reference.” The point of reference may be defined as at or associated with a physical object in the physical environment, for example a human subject **102** and/or a mobile device **104** through which a user (in this case human subject **102**) may provide control input. The point of reference may also be another point in space which may be specified via the mobile device **104** by clicking on a location of interest on a map or image. For example, a user **102** viewing a live video feed from UAV **100** through a touch display of mobile device **104** may touch a point or select a displayed object to redefine the point of reference about which motion is defined. Further, the defined point of reference may be stationary (e.g. a building or physical marker) or may be in motion (for example a moving car). For example, if the point of reference is set to a moving car, then any motions by the UAV **100** may be made relative to the car. In other words, if the point of reference is set to be a car moving at 25 mph, then a UAV **100** in “hover” would actually match the speed of the car while maintaining a constant position/orientation relative to the car. If the UAV **100** received input to move 10 feet in one direction, it would again do so relative to the car's position/orientation at any given moment. A particular point of reference can be tracked using any of the aforementioned systems for localization and navigation. Specifically, in an embodiment, a point of reference can be tracked in a 3D map of the surrounding environment generated using visual inertial odometry with captured stereoscopic images of the surrounding environment.

(52) A relative coordinate system may simplify the motion calculations necessary to maneuver the UAV **100**. Further, controlled motions made relative to point of reference associated with a subject **102** or mobile device **104** may allow for more intuitive control of the UAV **100**.

(53) FIGS. 7A-7B illustrate at a high level how the motion of a UAV **100** may be estimated and/or defined according to different coordinate systems. In FIG. 7A, the motion of an UAV **100** may be estimated and/or defined according to a cylindrical polar coordinate system relative to a point of reference, for example human subject **102** or a mobile device **104** held operated by a user (in this example human subject **102**). In such a configuration, a position (and thereby motion) relative to a point of reference may be defined according to the cylindrical polar coordinate system. For example, at any given time, the position of UAV **100** may be defined at a coordinate  $(r, \theta, z)$  relative to a point of reference (in this example human subject **102**). Further, in some embodiments, generated control commands may cause the UAV **100** to move along the normal tangent to an axial direction  $z$ . In other words, a control command, may cause the UAV **100** to accelerate along basis directions  $\hat{e}_z$  and  $\hat{e}_\theta$ , with no acceleration in the basis direction  $\hat{e}_r$ . Accordingly, in response to the particular control command, UAV **100** may travel along an invisible cylinder at a constant radius  $R$  from user **102**. Similarly, control commands may cause UAV **100** to accelerate along basis direction  $\hat{e}_r$  while maintaining constant positions  $z$  and  $\theta$ .

(54) Similarly, as shown in FIG. 7B, the motion of an UAV **100** may be estimated and/or defined according to a spherical polar coordinate system relative to a point of reference, for example human subject **102** or a mobile device **104** held operated by a user (in this example human subject **102**). Similar to the example illustrated in FIG. 7A, at any given time, the position of UAV **100** may be defined at a coordinate  $(r, \theta, \phi)$  relative to a point of reference (in this example human subject **102**). Further, in some embodiments, in order to maintain a constant separation with the point of reference, generated control commands may cause the UAV **100** to move along basis directions  $\hat{e}_\phi$  and  $\hat{e}_\theta$ , with no acceleration in basis direction  $\hat{e}_r$ . Accordingly, in response to the control command, UAV **100** may travel along an invisible spherical plane at a constant radius  $R$  from a point of reference (e.g. human subject **102**). Similarly, in order to close separation, control commands may cause UAV **100** to accelerate along basis direction  $\hat{e}_r$ .

(55) Calculations for the motion of the UAV **100** in the above described control configurations may be accomplished using relative or absolute coordinate system of any type (Cartesian, polar, cylindrical, etc.), although motion calculations based on an absolute coordinate system may be more processor intensive than if made relative to point of reference (e.g. human subject **102** or mobile device **104**). The cylindrical and polar coordinate systems are used here for illustrative purposes to describe more clearly the way in which the UAV **100** may move relative to a reference point (e.g. the human subject **102** or mobile device **104**) using the above described techniques.

(56) According to some embodiments, calculation of maneuvers to be performed by the UAV **100** may include implementation of a feed-forward control scheme. For example, as the motion of UAV **100** is continually estimated relative to a subject in motion and an obstacle is detected that will impede a planned path of the UAV **100**, the planned path may be continually updated in order to avoid the obstacle. This will allow for smoother transitions between flight maneuvers.

(57) While in flight, the UAV **100** may capture images and or video using one or more on board image capture devices (e.g. and image capture device mounted to a hybrid mechanical-digital gimbal). In some embodiments, image capture may track the same point of reference used for calculating motion (e.g. a human subject **102**). Consider an example in which a human subject **102** is the point of reference. Here, the UAV **100** may maneuver around the human subject **102** in response to generated control commands. Similarly, while moving around the point of reference (i.e. the human subject **102**), the UAV **100** may adjust the orientation and/or processing of image capture device(s) (e.g. cameras) such that the point of reference (i.e. the human subject **102**) remains centered in the field of view of the image capture device(s). Image capture may be adjusted according to techniques previously described, for example, by using a mechanical and/or a hybrid mechanical-digital gimbal system linked to one or more image capture devices.

(58) Adjusting Image Capture to Satisfy Specified Criteria Related to Image Capture Quality

(59) FIG. **8A** illustrates an UAV **100** in autonomous flight through a physical environment **802** that may dynamically adjust image capture of a subject (e.g. human subject **102**) to satisfy a specified criterion related to a quality of image capture.

(60) As shown in the example illustrated at FIG. **8A**, UAV **100** is at current position/orientation (indicated by the solid line drawing of a quadcopter) and is in autonomous flight along a current planned flight path **804** towards a future position/orientation (indicated by the dotted line drawing of the quadcopter). While in flight, UAV may be capturing images (including video) of a subject in the physical environment. In the example shown in FIG. **8**, UAV **100** is capturing images of human subject **102** as indicated by the image capture field of view **810**. The subject **102** may also be in motion through the physical environment. For example, human subject **102** is shown at a current position/orientation (indicated by the solid drawing of a human subject) and is in motion towards a future position/orientation (indicated by the dotted line drawing of a human subject). As previously described, image capture of the subject may be by one or more image capture devices (e.g. cameras) mounted to UAV **100**. In some embodiments, the image capture device(s) may be mounted to a motorized gimbal to enable visual stabilization and/or tracking. As previously described, in some embodiments, the motorized gimbal may be part of a hybrid mechanical-digital gimbal system.

(61) FIG. **8B** is flow chart of an example process **800b** for adjusting image capture by a UAV **100** to satisfy a specified criterion. Process **800b** begins at steps **802b** and **804b** with estimating a motion of the UAV **100** in autonomous flight through a physical environment **820** and estimating a motion of a subject through the physical environment **820**. The motion of UAV **100** and subject **102** may be estimated using any of the previously described techniques. For example, in some embodiments, the motions of the UAV **100** and/or subject **102** is estimated based in part on images captured by an image capture device associated with UAV **100**. As previously described, the image capture device of UAV **100** may include an array of multiple image capture device (e.g. cameras) providing an view of the surrounding physical environment (including subject **102**). In some



embodiments the motions of UAV **100** and/or subject **102** may be estimated using a process of visual inertial odometry by combining the captured images with sensor data from one or more inertial measurement units (IMU) onboard the UAV **100** and/or a mobile device **104** held by a human subject **102**. It shall be understood that the process of estimating motions may include estimating and tracking position/orientation of the UAV **100** and/or subject **102** over time. Further, it shall be understood that at any given point in time the UAV and or subject may be stationary (i.e. the estimated motion may effectively be zero).

(62) Process **800b** continues at step **806b** with in response to estimating the motions of the UAV **100** and the subject **102**, generating control commands to dynamically adjust image capture of the subject **102** by the image capture device associated with the UAV **100** to satisfy a specified criterion related to a quality of the image capture. In some embodiments control commands are generated by a flight controller onboard the UAV **100** (for example flight controller **1308** described with respect to FIG. **13**). Alternatively, the control commands may be generated by a remote computing device (e.g. a mobile device **104** or any other network-connected computing device) and transmitted to the UAV **100** via a wireless communication link (e.g. WiFi). The one or more specified criteria may be stored as instructions on a memory unit onboard the UAV **100** and/or in a remote computing device (e.g. a mobile device **104**). As used in this disclosure, the dynamic adjustment of image capture may include adjusting the characteristics or settings of an image capture device associated with UAV **100**, adjusting the orientation of an image capture device associated with UAV **100**, adjusting image processing (in real time or post processing), and or adjusting the motion of UAV **100** through the physical environment. For example, in some embodiments the control commands to dynamically adjust image capture may include control commands to adjust any of focus, aperture, shutter speed, light sensitivity (ISO), frame rate, color balance, image framing, or image stabilization by an image capture device associated with UAV **100**. As another example, in some embodiments the control commands to dynamically adjust image capture may include control commands to adjust subject tracking and image stabilization using an image capture device mounted to a motorized gimbal, for example a hybrid mechanical-digital gimbal. Further, in some embodiments, the control commands to dynamically adjust image capture may include control commands to adjust the position and/or orientation of UAV **100**.

(63) The terms “dynamic” or “dynamically” in this context imply that adjustments are made continually or near continuously as additional data is gathered (e.g. via an array of image capture devices mounted to UAV **100**). For example, the subject's motion may continually change, physical objects in the surrounding physical environment may be in motion and present obstacles, characteristics of the physical environment (e.g. weather, lighting, etc.) may change, and or the system may receive direct control inputs from a user. In some embodiments, motion (of the UAV **100** and subject **102**) is estimated and a planned flight path and image capture setting for UAV **100** is updated at period time increments in response to new data gathered by one or more sensors (e.g. image capture devices) associated with UAV **100**. These increments may be static (e.g. every 1 ms) or may, themselves, dynamically change. For example, in some embodiments, detected changes in the surrounding environment may cue a navigation/localization system to narrow the periodic increments at which motion estimations and image capture adjustments are made to more accurately respond to such changes in the surrounding environment. Alternatively, where the surrounding environment remains relatively static, a navigation/localization system may expand the periodic increments at which motion estimations and image capture adjustments are made so as to conserve power and/or data processing resources.

(64) According to some embodiments, a system in accordance with the present teachings may incorporate additional data when generating control commands to adjust image capture by an UAV **100**. For example, the generated control commands may further be based on data such as the geographical location of UAV **100** at the time of capture, the date and time images are captured, etc. For example, a system in accordance with the present teachings may determine (using any of

the aforementioned systems for localization) that UAV **100** is capturing images outdoors in the northern hemisphere in the evening during a summer month and may, accordingly adjust generated control commands to better suit those conditions. Similarly, a system in accordance with the present teachings may incorporate data gathered from other sensors associated with UAV **100**. For example, an ambient light sensor may gather data that may directly inform a level of ambient light in the physical environment without the need for additional processing of captured images.

(65) As previously described, in response to estimating the motions of the UAV **100** and the subject **102**, a computing system (e.g. a flight controller associated with UAV **100**) may generate control commands to dynamically adjust image capture to satisfy a specified criterion related to a quality of the image capture. It is generally understood that the quality of image capture in any given situation can depend on a number of different factors. For example, if the image capture is of a particular subject (e.g. a human, an animal, a vehicle, a building, or any other object), a basic determination on the quality of image capture may be whether the subject remains in view, in focus, properly framed, etc. Of course, this determination of what produces a quality image can depend on a number of objective and/or subject image capture quality standards. As described herein, a user may set one or more specified criteria related to image capture quality and based on those one or more criteria, a system in accordance with the present teaching may automatically generate control commands configured to dynamically adjust image capture by a UAV **100** to meet those specified one or more criteria without any direct control input by a user (e.g. remote control).

(66) As mentioned, the specified one or more criteria may be based on subjective and/or objective image capture quality standards. For example, a common objective standard for determining a quality of image capture is the balance of tonal distribution within the captured image. Such distribution can be graphically represented in the form of an image histogram. In an image histogram, the horizontal axis of the graph represents a range of tonal distribution, while the vertical axis represents the number of pixels at that particular tone. What results is a chart conveying information regarding the tonal variation in a given set of image data. For example, an image histogram, having most of its data points on the left side, may indicate a dark underexposed image. Conversely, an image histogram in which most of the data points fall on the right side may indicate a bright and perhaps overexposed image. A more balanced image histogram may therefore indicate a spread of tonal variation indicating greater contrast and therefore a higher quality image. The above example serves only to illustrate that the quality of an image may be based at least in part on certain objective criteria. Some other example of objective standards may be based on the variation in motion of a given shot indicating a steady or unsteady capture, balance of colors in a given shot, color saturation, variation in luminance within a given shot perhaps indicating an optimal or sub-optimal lighting source configuration (e.g. a shot of a subject individual with the sun at their back), focus of the shot, optical flow or any other standards tending to indicate an objective quality of a captured image.

(67) In some embodiments, the specified one or more criteria may be based on subjective image capture quality standards. It will be appreciated that the determined quality of a captured images may often be a matter of individual human perception. For example, aesthetic factors such as the framing of a shot, the sequencing of multiple shots, the orientation of a subject within a shot, or the background objects that share the shot with the subject may be perceived differently from one human to the next as indicating a higher quality or lower quality shot. This presents an issue when specifying criteria upon which to base control commands configured to adjust image capture by a UAV **100**. To address this issue, in some embodiments, subjective image quality standards may be based at least in part on polling and/or statistical methods applied to historical data to arrive at subjective standards that approximate generally held views of image quality.

(68) As an illustrative example, based on historical data (e.g. user reviews of sets of captured images) it may be determined that framing a subject within a shot according to the so called “golden ratio” (i.e. 1 to 1.618) produces an image of subjectively higher quality than an image with

a perfectly centered subject. With this in mind, a criterion may be specified that is related to this golden ratio-based subjective standard of image quality. For example, in response to estimating the motions of UAV **100** and a subject **102**, a system in accordance with the present teachings may generate control commands configured to adjust image capture (including maneuvering the UAV **100** relative to subject **102**) so as to achieve a composition roughly in line with the golden ratio thereby satisfying the specified criterion.

(69) According to some embodiments, subjective image quality standards may be based on existing statistical data. For example, a poll may be held in advance amongst a set of leading cinematographers or photographers. Their subjective input into the components of a high-quality image may then inform the standards upon which the one or more criteria are based. Such an implementation would in essence place a virtual expert cinematographer/photographer in control of the image capture capabilities of an UAV **100**.

(70) In some embodiments, subjective image quality standards may be based on continually gathered data associated with image quality and thereby continually evolve, for example by using machine learning algorithms. Consider the following example; multiple UAVs located around the world, may capture images that are then uploaded to a third-party social networking platform. The multiple users, forming a community on the social networking platform, may view each other's uploaded images and provide feedback (e.g. in the form of comments, upvotes, likes, etc.). Such a system may then aggregate the community feedback data with analyses of the uploaded images/video in order to inform and adjust the specified one or more criteria by which image capture adjustments are made for a given UAV **100**. As mentioned, machine learning algorithms may be applied to the gathered community feedback data to define certain subjective image quality standards.

(71) Returning to FIG. **8A**, in the described context an objective image capture quality standard may be (as previously described) to keep the subject in the field of view and in focus. For example, consider again the scenario illustrated in FIG. **8A**. A UAV **100** at a current position (as indicated by solid line quadcopter) is in autonomous flight a human subject **102** located in a physical environment **820**. Both UAV **100** and human subject **102** may be in motion or stationary. At a most basic level, the specified criterion in this context may simply be to keep the human subject **102** in view of an image capture device of UAV **100**. The specified criterion itself may include certain rules that must be met (within tolerance) to satisfy the criterion. For example, in the case of specified criterion to keep the subject “in view,” the criterion may be met as long as the image capture device captures the subject while the UAV **100** is no more than 100 meters from the subject. Similarly, set tolerances may define permissible limits non-adherence to these rules. For example, a set tolerance may be based on a temporal aspect (e.g. the subject **102** can fall out of view for no more than 2 seconds). In response to estimating motions of the UAV **100** and subject **102**, a system (e.g. a flight controller) may generate commands configured to adjust image capture by the UAV **100** to keep the human subject **102** in view. For example, as shown in FIG. **8A**, human subject **102** is in motion to a future location (as indicated by the dotted line human subject). Here, given the current motions of UAV **100** and human subject **102**, the control commands configured to keep the subject in view may include control commands to autonomously fly along a planned flight path **804**. Again, planned flight path **804** represents an instantaneous planned flight path given the current sensor data available. At a subsequent time step, this planned flight path may be adjusted (e.g. based on changes in the motion of the UAV **100** and/or subject **102** or the introduction of another physical obstacle) to satisfy the specified criterion of keeping the subject **102** in view.

(72) In some embodiments, motion planning for travel along a planned flight path (e.g. flight path **804**) may involve a technique generally referred to as “feed-forward” control. In other words, control commands may be generated (and continually updated) in anticipation of maneuvers needed to maintain autonomous flight along the current planned flight path. For example, consider planned flight path **804** shown in FIG. **8A**. From the current position of UAV **100** (e.g. as indicated

by the solid line quadcopter), a system in accordance with the present teachings may generate control commands that anticipate the various maneuvers needed to fly along the sweeping arc of flight path **804** and decelerate to arrive at a stopping position (e.g. future position indicated by the dotted line quadcopter). As mentioned, planned flight path **804** represents the current planned flight path at a given time. The planned flight path **804** may be continually adjusted (or at least adjusted at time intervals) to adjust image capture to satisfy a specified criterion.

(73) Given the idealized physical environment **820** illustrated in FIG. **8A** and assuming that the only specified criterion is to keep the human subject **102** in view, the system generating the control commands may have multiple options for dynamically adjusting image capture by the UAV **100** to meet the specified criterion. For example, control commands may be generated that cause UAV **100** to simply follow human subject **102** at a constant distance (or at least within a maximum separation distance) while maintaining a constant altitude (or at least above a minimum altitude).

Alternatively, control commands may be generated that cause UAV **100** to fly past human subject **102** while an image capture device configured for active tracking (e.g. using a hybrid mechanical-digital gimbal) is adjusted to keeps the subject **102** in the field of view **810**, as illustrated in FIG. **8A**. As will be described, the specified one or more criteria may include further constraints.

(74) In some embodiments, a criterion may be specified to keep the subject in view while avoiding a collision with another object in the physical environment. FIG. **9** shows an example scenario involving a UAV **100** in flight over a physical environment **920** and capturing images of a human subject **102**. As shown in FIG. **9**, UAV **100** may be in autonomous flight along a current planned flight path **904** to maneuver to avoid a collision with another object **930** in the physical environment while keeping human subject **102** in view (as indicated by field of view lines **910**). The example illustrated in FIG. **9** is idealized and shows a relatively large stationary object **930** (for example a building or other structure), but the same concepts may apply to avoid smaller mobile objects such as a bird in flight. As shown in FIG. **9**, based on the estimated motions of UAV **100** and subject **102**, a system in accordance with the present teachings may generate control commands to dynamically adjust image capture includes by generating control commands to maneuver UAV **100** along flight path **904** to avoid object **930** while keeping human subject in view (as indicated by field of view lines **910**). Notably, this illustrates that the addition of another constraint to the specified criterion (i.e. avoiding a collision) narrows the number of possible flight paths UAV **100** can take while still satisfying the specified criterion. For example, because the human subject **102** is moving to the right of object **930** and based on the characteristics of object **930**, in order to keep human subject **102** in view, UAV **100** must also maneuver to the right of object **930**. Any of the previously described localization techniques may be utilized to detect the presence of the object **930** with relation to human subject **102** and/or UAV **100** and to generate control commands configured to cause UAV **100** to avoid a collision with object **930**. For example, in some embodiments, based in part on images captured by an array of image capture devices mounted to UAV **100** and using a process of visual inertial odometry, the geometry and position/orientation of object **930** relative to UAV **100** may be determined.

(75) In some embodiments, a criterion may be specified to keep the subject in view and the image captured image unobstructed by another object in the physical environment. Similar to FIG. **9**, FIG. **10** shows an example scenario involving a UAV **100** in flight over a physical environment **1020** and capturing images of a human subject **102**. As shown in FIG. **10**, at a current time, human subject **102** is located on an opposite side of object **1010** from UAV **100** but is within a field of view of UAV **100** as indicated by the line of sight **1010** connecting the solid line UAV **100** to the solid line subject **102**. Based on the estimated motion of human subject **102** (as indicated by the arrow pointing away from human subject **102**), it may be determined that at a future time, human subject **102** may be located on the opposite side of object **1030** and not in view of UAV **100** (assuming UAV **100** remains stationary) as indicated by the obstructed line of sight line **1012** connecting the solid line UAV **100** to the dotted line subject **102**. Based on this predicted

obstruction, a system in accordance with the present teachings may generate control commands configured to adjust image capture to keep subject **102** in view and unobstructed in order to satisfy the specified criterion. Here, the generated control commands may include control commands configured to cause UAV **100** to maneuver along a flight path **1006** to keep the view of human subject **102** unobstructed. Note that in this example, simply avoiding a collision with object **1030** may not be sufficient to satisfy the specified criterion. For example, if the generated control commands cause UAV **100** to maneuver along alternative flight path **1004** instead of **1006**, its view of human subject **102** will become momentarily obstructed by object **1030**.

(76) As with FIG. **9**, the example illustrated in FIG. **10** is idealized and shows a relatively simple stationary object **1030** (for example a building or other structure). In this example, a specified criterion to avoid collision with the object may produce the same or similar results as a specified criterion to keep the view unobstructed. In other words, in both situations, UAV **100** may maintain a maximum separation distance while maneuvering to both avoid a collision with an object and keep the view unobstructed. However, consider an example with an object having more complex features such as a tree with sparse canopy cover. Here, if a criterion is specified to keep the subject in view while avoiding contact with the tree, a system in accordance with the present teachings may generate control commands configured to cause UAV **100** to fly over the tree while tracking the human subject **102** walking under the tree. Collision is avoided and because the canopy is sparse, the subject **102** remains in view. However, this will still result in poor image capture because the view of the subject **102** will be obstructed by intermittent leaves. Instead, if a criterion is specified to keep the view of the subject unobstructed, the system may instead generate control commands configured to cause UAV **100** to rapidly reduce its altitude to drop below the canopy of the tree and to continue to track human subject **102**. In such an example, UAV **100** may increase its altitude again once human subject **102** has emerged from under the canopy of the tree.

(77) As previously discussed, in some embodiments, the estimated motion of UAV **100** and subject **102** may be based in part on localization data relative to a computer-generated 3D map. For example if a pre-generated 3D map of the surrounding physical environment is available, the motions of UAV **100** and/or subject **102** relative to the 3D map may be estimated using any of the previously described localization techniques. Alternatively, if a pre-generated 3D map is not available, systems in accordance with the present teachings may continually generate and update a 3D map of the physical environment while the UAV **100** is in flight through the environment through a process sometimes referred to as SLAM (simultaneous localization and mapping). Again, as previously discussed, such a 3D map may be generated using a process of visual inertial odometry based in part on images captured by an image capture device associated with UAV **100**.

(78) In embodiments in which the motions of UAV **100** and subject **102** are estimated within a representative virtual 3D map of the physical environment, a system in accordance with the present teaching may define, within the 3D map, a virtual line between a virtual representation of an estimated position of the subject **102** and a virtual representation of an estimated position of the UAV **100**. Here, if a criterion is specified to keep the view of subject **102** unobstructed by other objects, the system may generate control commands configured to cause UAV **100** to fly a path such that the virtual line does not intersect a virtual representation of another physical object. Again, this criterion may be specified with a certain tolerance to account for objects in motion. In other words, if UAV **100** and/or subject **102** are both in motion, it may be inevitable that at certain times the virtual line connecting their representations in the virtual map may intersect representations of other objects. However, if that intersection persists for more than a certain period of time (e.g. 1 ms), the system may respond by generating control commands configured to cause UAV **100** to maneuver to avoid the intersection. Consider again the example scenario illustrated in FIG. **10**. Sight line **1010** may represent the virtual line connecting the representations of UAV **100** and subject **102** within a virtual environment (i.e. representative of physical environment **1020**). As human subject **102** begins to move within the physical environment (as shown by the arrow in FIG.

**10)**, the virtual line **100** connecting the virtual representations moves as well. Eventually, as human subject **102** moves behind object **1030**, the virtual line **1010** within the 3D map intersects the corner of a virtual representation of physical object **1030**. This intersection thereby causes the system to generate control commands configured to cause UAV **100** to maneuver to avoid the intersection. For example, the motion of the virtual line can be tracked and it may be determined that in order to avoid the intersection, UAV **100** should maneuver along flight path **1006** as opposed to flight path **1004** to keep the view of subject **102** unobstructed. This above described method also works to avoid obstruction by more complex objects. Consider again the example of a tree with sparse canopy cover. Here, the generated 3D map may be a voxel map configured to approximate the more complex objects in the physical environment. For example, virtual representation of the tree may include a number of virtual volumetric elements (i.e. voxels) representing the branches and leaves.

(79) At any given time, the virtual line may intersect with multiple virtual representations of objects, particularly in real world situations with multiple complex objects populating the physical environment. In other words, multiple points of intersection can be detected along the virtual line at a given time. Accordingly, to maintain line of sight, a flight path can be calculated to avoid visual obstruction by the multiple intersecting objects. This flight path can be relatively simple (for example similar to the arc of flight path **1006**) to avoid clusters of multiple objects (e.g. the leaves forming the canopy cover of a tree) or may include multiple complex maneuver intended to avoid visual obstruction caused by the multiple objects indicated by the multiple intersection points (e.g. multiple trees between a UAV **100** and subject **102**).

(80) In some situations, intersection points along a virtual line can be analyzed differently depending on their distance from UAV **100**. Motion by a UAV **100** generally has a greater impact on resolving visual obstruction caused by objects that are closer to the UAV **100**. This of course depends on the size and/or shape of the obstructing object, however in general relatively minor maneuvers by UAV **100** may be sufficient to maintain line of sight with a subject around an object that is close to UAV **100**. Conversely, more drastic maneuvers by UAV **100** may be necessary to maintain line of sight around an object that is closer to subject **102**. This makes sense when again considering the scenario described in FIG. **10**. Although described as a single object **1030**, the virtual representation of object **1030** can also be described as multiple surfaces that intersect the virtual line at multiple points. For example, obstructed line of sight line **1012** intersects a first surface of object **1030** that faces UAV **100** at a first point and a second surface of object **1030** that faces a future position of subject **102** at a second point. A minor maneuver along flight path **1006** may be sufficient such that sight line **1012** no longer intersects the first surface (i.e. the surface closest to UAV **100**) at the first point. However, a more extended maneuver along flight path **1006** may be necessary before sight line **1012** no longer intersects the second surface (i.e. the surface closest to subject **102**) at the second point, thereby establishing line of sight with subject **102**.

(81) In some embodiments a specified criterion (e.g. to maintain line of sight with a subject) can be applied or combined with inputs by a user. For example, consider a human user operating controlling UAV **100** using a device **104**. The user inputs control commands via device **104** that are transmitted to UAV **100** causing it to maneuver through the physical environment in much the same way a remote control system operates. However, a control criterion may be specified and applied to any input control commands to maintain line of sight with a subject **102** (e.g. the user). In other words, UAV **100** may maneuver according to the user's direct control commands but may adjust certain maneuvers to maintain line of sight with a subject despite these control commands. In some embodiments, the specified criterion can be applied to modify direct control commands from a user. Alternatively, user inputs can be combined with motion estimations of the UAV **100** and/or subject **102** to generate control commands that closely approximate maneuvers intended by the user input while satisfying a given specified criterion (e.g. line of sight).

(82) This application of specified criterion such as maintaining line of sight to user inputs can be

particularly useful where certain flight regulations are enforced. For example, the Federal Aviation Administration (FAA) and Department of Transportation (DOT) are currently in the process of finalizing flight regulations applicable to certain UAVs that require that a remote operator (i.e. a user) maintain visual contact with the UAV at all times. A simple control restraint on separation distance may be helpful to an extent in enforcing a line of sight regulation. For example, a UAV can be configured so that it is never outside a maximum separation distance from an associated control device. However, such a constraint will not account for situations in which line of sight is obstructed by objects within the maximum separation distance. Instead, using the aforementioned techniques, line of sight regulations may be more effectively and automatically enforced.

(83) In some embodiments a criterion may be specified to adjust image capture according to a predefined script to suit a particular type of scene being captured. For example, in some embodiments, in response to the determined motions of the UAV **100** and subject **102**, a system may generate control commands configured to cause the UAV **100** to fly a pre-scripted flight path and/or capture images using predefined settings. Here the pre-scripted flight path and/or image capture settings may be based on one or more objective and/or subject image quality standards. As an illustrative example, it may be subjectively desirable to perform certain pre-scripted flyover shots where the subject being captured is part of a scene with a high level of action.

(84) Consider the example scenario illustrated in FIG. **11**. As shown in FIG. **11**, a UAV **100** is in autonomous flight over a physical environment **1120** while tracking and capturing images of a human subject **102**. Here, human subject **102** is shown as a skier traveling down a mountainside. Based on a specified criterion, and in response to estimating the motions of UAV **100** and subject **102**, control commands may be generated that include control commands configured to cause UAV **100** to fly pre-scripted flyover path **1104** over human subject **102** to capture subjectively high-quality action-based video of human subject **102**. Again, adjustments in image capture may be made dynamically in response to the changing motions of UAV **100** and/or subject **102**. In the scenario illustrated in FIG. **11**, consider a situation in which after having flown pre-scripted flight path **1104**, the subject **102** then skis off a jump and performs an acrobatic maneuver in mid-air. In response to the newly estimated motions of the UAV **100** and subject **102**, and in order to satisfy the specified criterion, control commands may be generated that are configured to cause UAV **100** to fly an alternative pre-scripted flight path (not shown). For example, in order to capture the action of the mid-air maneuver by the human subject **102**, control commands may be generated to cause UAV **102** to quickly fly under subject **102** while tracking image capture of subject **102**.

Alternatively, or in addition to flying according to a pre-scripted flight path, control commands may be generated that adjust certain characteristics or settings of the image capture device to deliver subjectively higher quality image capture. For example, using the scenario illustrated in FIG. **11**, subjective image quality standards for capturing downhill skiing scenes may dictate that the image capture device be set to, for example, a certain focal length, shutter speed, white balance, exposure, etc.

(85) In some embodiments, in addition to estimating the motions of UAV **100** and subject **102**, a system in accordance with the present teachings may automatically determine (based in part on those estimated motions) the type of scene being captured. For example, a process for determining the type of scene being captured may include first identifying the subject **102** in the physical environment. In the aforementioned embodiments, the subject **102** is described as a general object (human or otherwise) that is the focus of image capture by UAV **100**. Tracking of a subject **102** may require differentiating the subject as a discrete object apart from its surroundings, but does not necessarily require identifying what the subject **102** is. In some embodiments, this process of identifying the subject **102** may include comparing (in real time or near real time) captured images of the subject **102** against stored 2D and/or 3D appearance models to determine the type of object subject **102** most closely matches. The level of categorization can depend here. For example, it may be enough to simply identify the subject as human. Alternatively, the subject can be identified as

particular type of human subject (e.g. a skier), or even as a particular individual (e.g. champion downhill skier, John Doe).

(86) Once the subject is identified, the process may continue by determining based on the identification of the subject **102** and the estimated motion of the subject **102** that the image captured by UAV **100** of the subject **102** is intended for a particular type of scene. Note that this process of determining that image capture is intended for a particular type of scene may also involve first identifying other objects in the physical environment. Again, consider the scenario described with respect to FIG. **11**. Based on a comparison of captured images of subject **102** against stored appearance models, a system may identify subject **102** as a skier. Further, based on a comparison of captured images of other objects against stored appearance models, the system may identify those other objects as, for example, other skiers, snowcapped mountains, trees, etc. Based at least in part on the identification of the subject as a skier, an estimated motion of the subject, and perhaps identification of other objects (e.g. mountains, trees, etc.), the system may determine that the scene being captured is a downhill ski scene.

(87) Having determined that the scene being captured is a particular type of scene (e.g. a downhill ski scene), and in order to satisfy the specified criterion, the system may generate control commands configured to cause UAV **100** to fly a pre-scripted flight and/or adjust certain characteristics or settings of the image capture device based on the particular type of scene.

(88) In some embodiments, a criterion may be specified to keep avoid backlighting of the captured subject. Consider the example scenario illustrated in FIG. **12**. As shown in FIG. **12**, a UAV **100** is in autonomous flight over a physical environment **1220** while tracking and capturing images of a human subject **102**. The human subject **102** is lit by a light source **1250** from one side, in this example the Sun. Here, UAV **100** is shown at a current location (as indicated by the solid line quadcopter) opposite the light source **1250** relative to the human subject **102**. At this current position, images capture of human subject **102** (for example within field of view **1210**) are likely to be devoid of much detail of human subject **102** due to the shadow cast by the light source **1250**. In the case of a powerful light source **1250** such as the Sun, the captured images may be completely washed out due to over exposure, particularly if the image capture device associated with UAV **100** is oriented so as to be pointed substantially in the direction of the light source **1250**. While perhaps based on a subjective standard, backlight image capture such as this is generally understood to result in poor quality images.

(89) According, in some embodiments, a criterion specified to avoid backlighting, a system in accordance with the present teachings may generate control commands configured such that, at a given time, the UAV **100** is positioned substantially between the light source and the subject **120** so as to avoid backlighting in the images captured of the subject **102**. A method for generating such command may include, in addition to estimating the motions of UAV **100** and subject **102**, also estimating a position of a light source. This may be accomplished in a number of ways for example, by processing images captured by an image capture device associated with UAV **100** and/or based on localization data of known light sources (e.g. the Sun). Given global positioning information for UAV **100** and the current date/time, a localization system can determine if UAV **100** is pointed towards the Sun while capturing images of a subject **102**.

(90) Consider again the example scenario illustrated in FIG. **12**. Here, because UAV **100** is located opposite a major light source **1250** while capturing images of subject **102**, in order to satisfy the specified criterion, a system in accordance with the present teachings may generate control commands configured to cause UAV **100** to autonomously maneuver along flight path **1204** until, at a future time, UAV **100** is located substantially between light source **1250** and subject **102** (as indicated by the dotted line quadcopter).

(91) Unmanned Aerial Vehicle—Example System

(92) A UAV **100**, according to the present teachings, may be implemented as any type of unmanned aerial vehicle. An Unmanned Aerial Vehicle (UAV), sometimes referred to as a drone, is generally



defined as any aircraft capable of controlled flight without a human pilot onboard. UAVs may be controlled autonomously by onboard computer processors or via remote control by a remotely located human pilot. Similar to an airplane, UAVs may utilize fixed aerodynamic surfaces along means for propulsion (e.g. propeller, jet) to achieve lift. Alternatively, similar to helicopters, UAVs may directly use the means for propulsion (e.g. propeller, jet, etc.) to counter gravitational forces and achieve lift. Propulsion-driven lift (as in the case of helicopters) offers significant advantages in certain implementations, for example as a mobile filming platform, because it allows for controlled motion along all axis.

(93) Multi-rotor helicopters, in particular quadcopters, have emerged as a popular UAV configuration. A quadcopter (also known as a quadrotor helicopter or quadrotor) is a multirotor helicopter that is lifted and propelled by four rotors. Unlike most helicopters, quadcopters use two sets of two fixed-pitch propellers. A first set of rotors turns clockwise, while a second set of rotors turns counter-clockwise. In turning opposite directions, the first set of rotors may counter the angular torque caused by the rotation of the other set, thereby stabilizing flight. Flight control is achieved through variation in the angular velocity of each of the four fixed-pitch rotors. By varying the angular velocity of each of the rotors, a quadcopter may perform precise adjustments in its position (e.g. adjustments in altitude and level flight left, right, forward and backward) and orientation, including pitch (rotation about a first lateral axis), roll (rotation about a second lateral axis), and yaw (rotation about a vertical axis). For example, if all four rotors are spinning (two clockwise, and two counter-clockwise) at the same angular velocity, the net aerodynamic torque about the vertical yaw axis is zero. Provided the four rotors spin at sufficient angular velocity to provide a vertical thrust equal to the force of gravity, the quadcopter can maintain a hover. An adjustment in yaw may be induced by varying the angular velocity of a subset of the four rotors thereby mismatching the cumulative aerodynamic torque of the four rotors. Similarly, an adjustment in pitch and/or roll may be induced by varying the angular velocity of a subset of the four rotors but in a balanced fashion such that lift is increased on one side of the craft and decreased on the other side of the craft. An adjustment in altitude from hover may be induced by applying a balanced variation in all four rotors thereby increasing or decreasing the vertical thrust. Positional adjustments left, right, forward, and backward may be induced through combined pitch/roll maneuvers with balanced applied vertical thrust. For example, to move forward on a horizontal plane, the quadcopter would vary the angular velocity of a subset of its four rotors in order to perform a pitch forward maneuver. While pitching forward, the total vertical thrust may be increased by increasing the angular velocity of all the rotors. Due to the forward pitched orientation, the acceleration caused by the vertical thrust maneuver will have a horizontal component and will therefore accelerate the craft forward on horizontal plane.

(94) FIG. 13 shows a diagram of an example UAV system **1300** including various functional system components that may be part of a UAV **100**, according to some embodiments. UAV system **1300** may include one or more means for propulsion (e.g. rotors **1302** and motor(s) **1304**), one or more electronic speed controllers **1306**, a flight controller **1308**, a peripheral interface **1310**, a processor(s) **1312**, a memory controller **1314**, a memory **1316** (which may include one or more computer readable storage mediums), a power module **1318**, a GPS module **1320**, a communications interface **1322**, an audio circuitry **1324**, an accelerometer **1326** (including subcomponents such as gyroscopes), an inertial measurement unit (IMU) **1328**, a proximity sensor **1330**, an optical sensor controller **1332** and associated optical sensor(s) **1334**, a mobile device interface controller **1336** with associated interface device(s) **1338**, and any other input controllers **1340** and input device **1342**, for example display controllers with associated display device(s). These components may communicate over one or more communication buses or signal lines as represented by the arrows in FIG. 13. As mentioned earlier, piloting input may be provided wirelessly by a user **102** on the ground or in another vehicle via remote control or portable multi-function device **104**.

(95) UAV system **1300** is only one example of a system that may be part of a UAV **100**. A UAV **100** may include more or fewer components than shown in system **1300**, may combine two or more components as functional units, or a may have a different configuration or arrangement of the components. Some of the various components of system **1300** shown in FIG. **13** may be implemented in hardware, software or a combination of both hardware and software, including one or more signal processing and/or application specific integrated circuits. Also, UAV **100** may include an off-the-shelf UAV (e.g. a currently available remote-controlled quadcopter) coupled with a modular add-on device (for example one including components within outline **1390**) to perform the innovative functions described in this disclosure.

(96) As described earlier, the means for propulsion **1302-1304** may comprise a fixed-pitch rotor. The means for propulsion may also be a variable-pitch rotor (for example, using a gimbal mechanism), a variable-pitch jet engine, or any other mode of propulsion having the effect of providing force. The means for propulsion **1302-1304** may include a means for varying the applied thrust, for example via an electronic speed controller **1306** varying the speed of each fixed-pitch rotor.

(97) Flight Controller **1308** (sometimes referred to as a “flight control system” or “autopilot”) may include a combination of hardware and/or software configured to receive input data (e.g. sensor data from image capture devices **1334**), interpret the data and output control commands to the propulsion systems **1302-1306** and/or aerodynamic surfaces (e.g. fixed wing control surfaces) of the UAV **100**. Alternatively, or in addition, a flight controller **1308** may be configured to receive control commands generated by another component or device (e.g. processors **1312** and/or a separate computing device), interpret those control commands and generate control signals to the propulsion systems **1302-1306** and/or aerodynamic surfaces (e.g. fixed wing control surfaces) of the UAV **100**.

(98) Memory **1316** may include high-speed random-access memory and may also include non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Access to memory **1316** by other components of system **1300**, such as the processors **1312** and the peripherals interface **1310**, may be controlled by the memory controller **1314**.

(99) The peripherals interface **1310** may couple the input and output peripherals of system **1300** to the processor(s) **1312** and memory **1316**. The one or more processors **1312** run or execute various software programs and/or sets of instructions stored in memory **1316** to perform various functions for the UAV **100** and to process data. In some embodiments, processors **1312** may include general central processing units (CPUs), specialized processing units such as Graphical Processing Units (GPUs) particularly suited to parallel processing applications, or any combination thereof.

(100) In some embodiments, the peripherals interface **1310**, the processor(s) **1312**, and the memory controller **1314** may be implemented on a single integrated chip. In some other embodiments, they may be implemented on separate chips.

(101) The network communications interface **1322** may facilitate transmission and reception of communications signals often in the form of electromagnetic signals. The transmission and reception of electromagnetic communications signals may be carried out over physical media such copper wire cabling or fiber optic cabling, or may be carried out wirelessly for example, via a radiofrequency (RF) transceiver. In some embodiments the network communications interface may include RF circuitry. In such embodiments, RF circuitry may convert electrical signals to/from electromagnetic signals and communicate with communications networks and other communications devices via the electromagnetic signals. The RF circuitry may include well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. The RF circuitry may facilitate transmission and receipt of data over communications networks (including public,

private, local, and wide area). For example, communication may be over a wide area network (WAN), a local area network (LAN), or a network of networks such as the Internet. Communication may be facilitated over wired transmission media (e.g. via Ethernet) or wirelessly. Wireless communication may be over a wireless cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other modes of wireless communication. The wireless communication may use any of a plurality of communications standards, protocols and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11n and/or IEEE 802.11ac), voice over Internet Protocol (VoIP), Wi-MAX, or any other suitable communication protocols.

(102) The audio circuitry **1324**, including the speaker and microphone **1350** may provide an audio interface between the surrounding environment and the UAV **100**. The audio circuitry **1324** may receive audio data from the peripherals interface **1310**, convert the audio data to an electrical signal, and transmits the electrical signal to the speaker **1350**. The speaker **1350** may convert the electrical signal to human-audible sound waves. The audio circuitry **1324** may also receive electrical signals converted by the microphone **1350** from sound waves. The audio circuitry **1324** may convert the electrical signal to audio data and transmits the audio data to the peripherals interface **1310** for processing. Audio data may be retrieved from and/or transmitted to memory **1316** and/or the network communications interface **1322** by the peripherals interface **1310**.

(103) The I/O subsystem **1360** may couple input/output peripherals of UAV **100**, such as an optical sensor system **1334**, the mobile device interface **1338**, and other input/control devices **1342**, to the peripherals interface **1310**. The I/O subsystem **1360** may include an optical sensor controller **1332**, a mobile device interface controller **1336**, and other input controller(s) **1340** for other input or control devices. The one or more input controllers **1340** receive/send electrical signals from/to other input or control devices **1342**.

(104) The other input/control devices **1342** may include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, touch screen displays, slider switches, joysticks, click wheels, and so forth. A touch screen display may be used to implement virtual or soft buttons and one or more soft keyboards. A touch-sensitive touch screen display may provide an input interface and an output interface between the UAV **100** and a user. A display controller may receive and/or send electrical signals from/to the touch screen. The touch screen may display visual output to the user **102**. The visual output may include graphics, text, icons, video, and any combination thereof (collectively termed “graphics”). In some embodiments, some or all of the visual output may correspond to user-interface objects, further details of which are described below.

(105) A touch sensitive display system may have a touch-sensitive surface, sensor or set of sensors that accepts input from the user based on haptic and/or tactile contact. The touch sensitive display system and the display controller (along with any associated modules and/or sets of instructions in memory **1316**) may detect contact (and any movement or breaking of the contact) on the touch screen and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys or images) that are displayed on the touch screen. In an exemplary embodiment, a point of contact between a touch screen and the user corresponds to a finger of the user.

(106) The touch screen may use LCD (liquid crystal display) technology, or LPD (light emitting polymer display) technology, although other display technologies may be used in other embodiments. The touch screen and the display controller may detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with a touch screen.

(107) The mobile device interface device **1338** along with mobile device interface controller **1336** may facilitate the transmission of data between a UAV **100** and a mobile device **104** (for example, in use as a control device by a user). According to some embodiments, communications interface **1322** may facilitate the transmission of data between UAV **100** and a mobile device **104** (for example, where data is transferred over a local Wi-Fi network).

(108) UAV system **1300** also includes a power system **1318** for powering the various components. The power system **1318** may include a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in computerized device.

(109) UAV system **1300** may also include one or more image capture devices **1334**. FIG. **13** shows an image capture device **1334** coupled to an image capture controller **1332** in I/O subsystem **1360**. The image capture device **1334** may include one or more optical sensors. For example, image capture device **1334** may include a charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. The optical sensors of image capture device **1334** receive light from the environment, projected through one or more lens (the combination of an optical sensor and lens can be referred to as a “camera”) and converts the light to data representing an image. In conjunction with an imaging module located in memory **1316**, the image capture device **1334** may capture images (including still images and/or video). In some embodiments, an image capture device **1334** may include a single fixed camera. In other embodiments, an image capture device **1334** may include a single adjustable camera (adjustable using a gimbal mechanism with one or more axes of motion). In some embodiments, an image capture device **1334** may include a camera with a wide-angle lens providing a wider field of view. In some embodiments, an image capture device **1334** may include an array of multiple cameras providing up to a full 360 degree view in all directions. In some embodiments, an image capture device **1334** may include two or more cameras (of any type as described herein) placed next to each other in order to provide stereoscopic vision. In some embodiments, an image capture device **1334** may include multiple cameras of any combination as described above. For example, UAV **100** may include four sets of two cameras each positioned so as to provide a stereoscopic view at multiple angles around the UAV **100**. In some embodiments, a UAV **100** may include some cameras dedicated for image capture of a subject and other cameras dedicated for image capture for navigation (e.g. through visual inertial odometry).

(110) UAV system **1300** may also include one or more proximity sensors **1330**. FIG. **13** shows a proximity sensor **1330** coupled to the peripherals interface **1310**. Alternately, the proximity sensor **1330** may be coupled to an input controller **1340** in the I/O subsystem **1360**. Proximity sensors **1330** may generally include remote sensing technology for proximity detection, range measurement, target identification, etc. For example, proximity sensors **1330** may include radar, sonar, and light illuminated detection and ranging (Lidar).

(111) UAV system **1300** may also include one or more accelerometers **1326**. FIG. **13** shows an accelerometer **1326** coupled to the peripherals interface **1310**. Alternately, the accelerometer **1326** may be coupled to an input controller **1340** in the I/O subsystem **1360**.

(112) UAV system **1300** may include one or more inertial measurement units (IMU) **1328**. An IMU **1328** may measure and report the UAV's velocity, acceleration, orientation, and gravitational forces using a combination of gyroscopes and accelerometers (e.g. accelerometer **1326**).

(113) UAV system **1300** may include a global positioning system (GPS) receiver **1320**. FIG. **13** shows a GPS receiver **1320** coupled to the peripherals interface **1310**. Alternately, the GPS receiver **1320** may be coupled to an input controller **1340** in the I/O subsystem **1360**. The GPS receiver **1320** may receive signals from GPS satellites in orbit around the earth, calculate a distance to each of the GPS satellites (through the use of GPS software), and thereby pinpoint a current

global position of UAV **100**. In some embodiments, positioning of UAV **100** may be accomplished without GPS satellites through the use of other techniques as described herein.

(114) In some embodiments, the software components stored in memory **1316** may include an operating system, a communication module (or set of instructions), a flight control module (or set of instructions), a localization module (or set of instructions), a computer vision module, a graphics module (or set of instructions), and other applications (or sets of instructions). For clarity one or more modules and/or applications may not be shown in FIG. **13**.

(115) The operating system (e.g., Darwin, RTXC, LINUX, UNIX, OS X, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

(116) A communications module may facilitate communication with other devices over one or more external ports **1344** and may also include various software components for handling data transmission via the network communications interface **1322**. The external port **1344** (e.g., Universal Serial Bus (USB), FIREWIRE, etc.) may be adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.).

(117) A graphics module may include various software components for processing, rendering and displaying graphics data. As used herein, the term “graphics” may include any object that can be displayed to a user, including without limitation text, still images, videos, animations, icons (such as user-interface objects including soft keys), and the like. The graphics module in conjunction with a graphics processing unit (GPU) **1312** may process in real time or near real time, graphics data captured by optical sensor(s) **1334** and/or proximity sensors **1330**.

(118) A computer vision module, which may be a component of graphics module, provides analysis and recognition of graphics data. For example, while UAV **100** is in flight, the computer vision module along with graphics module (if separate), GPU **1312**, and image capture devices(s) **1334** and/or proximity sensors **1330** may recognize and track the captured image of a subject located on the ground. The computer vision module may further communicate with a localization/navigation module and flight control module to update a relative position between UAV **100** and a point of reference, for example a target subject (e.g. a mobile device or human subject), and provide course corrections to fly along a planned flight path relative to the point of reference.

(119) A localization/navigation module may determine the location and/or orientation of UAV **100** and provides this information for use in various modules and applications (e.g., to a flight control module in order to generate commands for use by the flight controller **1308**).

(120) Image capture devices(s) **1334** in conjunction with, image capture device controller **1332**, and a graphics module, may be used to capture images (including still images and video) and store them into memory **1316**.

(121) Each of the above identified modules and applications correspond to a set of instructions for performing one or more functions described above. These modules (i.e., sets of instructions) need not be implemented as separate software programs, procedures or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various embodiments. In some embodiments, memory **1316** may store a subset of the modules and data structures identified above. Furthermore, memory **1316** may store additional modules and data structures not described above.

(122) Example Computer Processing System

(123) FIG. **14** is a block diagram illustrating an example of a computer processing system **1400** in which at least some operations described herein can be implemented, consistent with various embodiments. Computer processing system **1400** can represent or be part of any of aforementioned devices (e.g. mobile device **104** or a general purpose computing device). Any of these systems can include two or more computer processing systems, as is represented in FIG. **14**, which can be

coupled to each other via a network or multiple networks.

(124) In the illustrated embodiment, the computer processing system **1400** includes one or more processors **1410**, memory **1411**, one or more communications devices **1412**, and one or more input/output (I/O) devices **1413**, all coupled to each other through an interconnect **1414**. The interconnect **1414** may be or include one or more conductive traces, buses, point-to-point connections, controllers, adapters and/or other conventional connection devices. The processor(s) **1410** may be or include, for example, one or more central processing units (CPU), graphical processing units (GPU), other general-purpose programmable microprocessors, microcontrollers, application specific integrated circuits (ASICs), programmable gate arrays, or the like, or any combination of such devices. The processor(s) **1410** control the overall operation of the computer processing system **1400**. Memory **1411** may be or include one or more physical storage devices, which may be in the form of random-access memory (RAM), read-only memory (ROM) (which may be erasable and programmable), flash memory, miniature hard disk drive, or other suitable type of storage device, or any combination of such devices. Memory **1411** may be or include one or more discrete memory units or devices. Memory **1411** can store data and instructions that configure the processor(s) **1410** to execute operations in accordance with the techniques described above. The communication device **1412** represents an interface through which computing system **1400** can communicate with one or more other computing systems. Communication device **712** may be or include, for example, an Ethernet adapter, cable modem, Wi-Fi adapter, cellular transceiver, Bluetooth transceiver, or the like, or any combination thereof. Depending on the specific nature and purpose of the computer processing system **1400**, the I/O device(s) **1413** can include various devices for input and output of information, e.g., a display (which may be a touch screen display), audio speaker, keyboard, mouse or other pointing device, microphone, camera, etc.

(125) Unless contrary to physical possibility, it is envisioned that (i) the methods/steps described above may be performed in any sequence and/or in any combination, and that (ii) the components of respective embodiments may be combined in any manner.

(126) The techniques introduced above can be implemented by programmable circuitry programmed/configured by software and/or firmware, or entirely by special-purpose circuitry, or by any combination of such forms. Such special-purpose circuitry (if any) can be in the form of, for example, one or more application-specific integrated circuits (ASICs), programmable logic devices (PLDs), field-programmable gate arrays (FPGAs), etc.

(127) Software or firmware to implement the techniques introduced here may be stored on a machine-readable storage medium and may be executed by one or more general-purpose or special-purpose programmable microprocessors. A “machine-readable medium”, as the term is used herein, includes any mechanism that can store information in a form accessible by a machine (a machine may be, for example, any computing device or system including elements similar to as described with respect to computer processing system **1400**). For example, a machine-accessible medium includes recordable/non-recordable media (e.g., read-only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory devices; etc.), etc.

#### OTHER REMARKS

(128) In this description, references to “an embodiment”, “one embodiment” or the like, mean that the particular feature, function, structure or characteristic being described is included in at least one embodiment of the technique introduced here. Occurrences of such phrases in this specification do not necessarily all refer to the same embodiment. Note that any and all of the embodiments described above can be combined with each other, except to the extent that it may be stated otherwise above or to the extent that any such embodiments might be mutually exclusive in function and/or structure.

(129) Although the disclosed technique has been described with reference to specific exemplary embodiments, it will be recognized that the technique is not limited to the embodiments described, but can be practiced with modification and alteration within scope of the appended claims.

Accordingly, the specification and drawings are to be regarded in an illustrative sense rather than a restrictive sense.

## Claims

1. A method comprising: capturing, by multiple image sensors of a vehicle, images of a physical environment surrounding the vehicle; processing the images of the physical environment surrounding the vehicle to identify images with overlapping fields of view; processing the images with overlapping fields of view to identify dense correspondences between the images, wherein the dense correspondences indicate matches between pixels in the images with overlapping fields of view; estimating distances to the pixels in the images with overlapping fields of view using the identified dense correspondences; and generating and continually updating a three-dimensional (3D) map of the physical environment based on the estimated distances.
2. The method of claim 1, further comprising: causing the vehicle to autonomously navigate through the physical environment based, at least in part, on the 3D map of the physical environment.
3. The method of claim 1, wherein the images of the physical environment surrounding the vehicle are captured during sequential time steps.
4. The method of claim 1, wherein the images of the physical environment surrounding the vehicle are stereoscopic images captured during a same timestep and the multiple image sensors are configured with pre-defined spatial offsets.
5. The method of claim 1, further comprising: estimating a relative position of the vehicle within the physical environment while simultaneously continually updating the 3D map.
6. The method of claim 1, further comprising: estimating an orientation of the vehicle within the physical environment while simultaneously continually updating the 3D map.
7. The method of claim 1, further comprising: estimating a relative position of a subject within the physical environment while simultaneously continually updating the 3D map.
8. The method of claim 1, further comprising: estimating an orientation of a subject within the physical environment while simultaneously continually updating the 3D map.
9. A vehicle comprising: multiple image sensors configured to capture images of a physical environment surrounding the vehicle; a control system configured to continually: process the images of the physical environment surrounding the vehicle to identify images with overlapping fields of view; process the images with overlapping fields of view to identify dense correspondences between the images, wherein the dense correspondences indicate matches between pixels in the images with overlapping fields of view; estimate distances to pixels in the images with the overlapping fields of view using the identified dense correspondences; and update a three-dimensional (3D) map of the physical environment based on the estimated distances.
10. The vehicle of claim 9, wherein the control system is further configured to: generate the 3D map of the physical environment based on the estimated distances.
11. The vehicle of claim 9, wherein the control system is further configured to: cause the vehicle to autonomously navigate through the physical environment based, at least in part, on the 3D map of the physical environment.
12. The vehicle of claim 9, wherein the images of the physical environment surrounding the vehicle are captured during sequential time steps.
13. The vehicle of claim 9, wherein the images of the physical environment surrounding the vehicle are stereoscopic images captured during a same timestep and the multiple image sensors are configured with pre-defined spatial offsets.
14. The vehicle of claim 9, wherein the control system is further configured to: estimate a relative position of the vehicle and/or a subject within the physical environment while simultaneously continually updating the 3D map.

15. The vehicle of claim 9, wherein the control system is further configured to: estimating an orientation of the vehicle and/or a subject within the physical environment while simultaneously continually updating the 3D map.

16. The vehicle of claim 9, wherein the multiple image sensors are arranged to provide a 360 degree view around the vehicle, and wherein the multiple image sensors are arranged such that at least two image sensors are provided with overlapping fields of view.

17. An apparatus, comprising: one or more memory units storing instructions that, when executed by one or more processors of a vehicle, cause the one or more processors to: process images of a physical environment surrounding the vehicle to identify images with overlapping fields of view; process the images with overlapping fields of view to identify dense correspondences between the images, wherein the dense correspondences indicate matches between pixels in the images with overlapping fields of view; estimate distances to pixels in the images with the overlapping fields of view using the identified dense correspondences; and generate and continually update a three-dimensional (3D) map of the physical environment based on the estimated distances.

18. The apparatus of claim 17, wherein the instructions, when executed by the one or more processors of the vehicle, further cause the one or more processors to: cause the vehicle to autonomously navigate through the physical environment based, at least in part, on the 3D map of the physical environment.

19. The apparatus of claim 17, wherein the instructions, when executed by the one or more processors of the vehicle, further cause the one or more processors to: estimate a relative position of the vehicle and/or a subject within the physical environment while simultaneously continually updating the 3D map.

20. The apparatus of claim 17, wherein the instructions, when executed by the one or more processors of the vehicle, further cause the one or more processors to: estimate an orientation of the vehicle and/or a subject within the physical environment while simultaneously continually updating the 3D map.

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